AI EVOLUTION

QUARTERLY

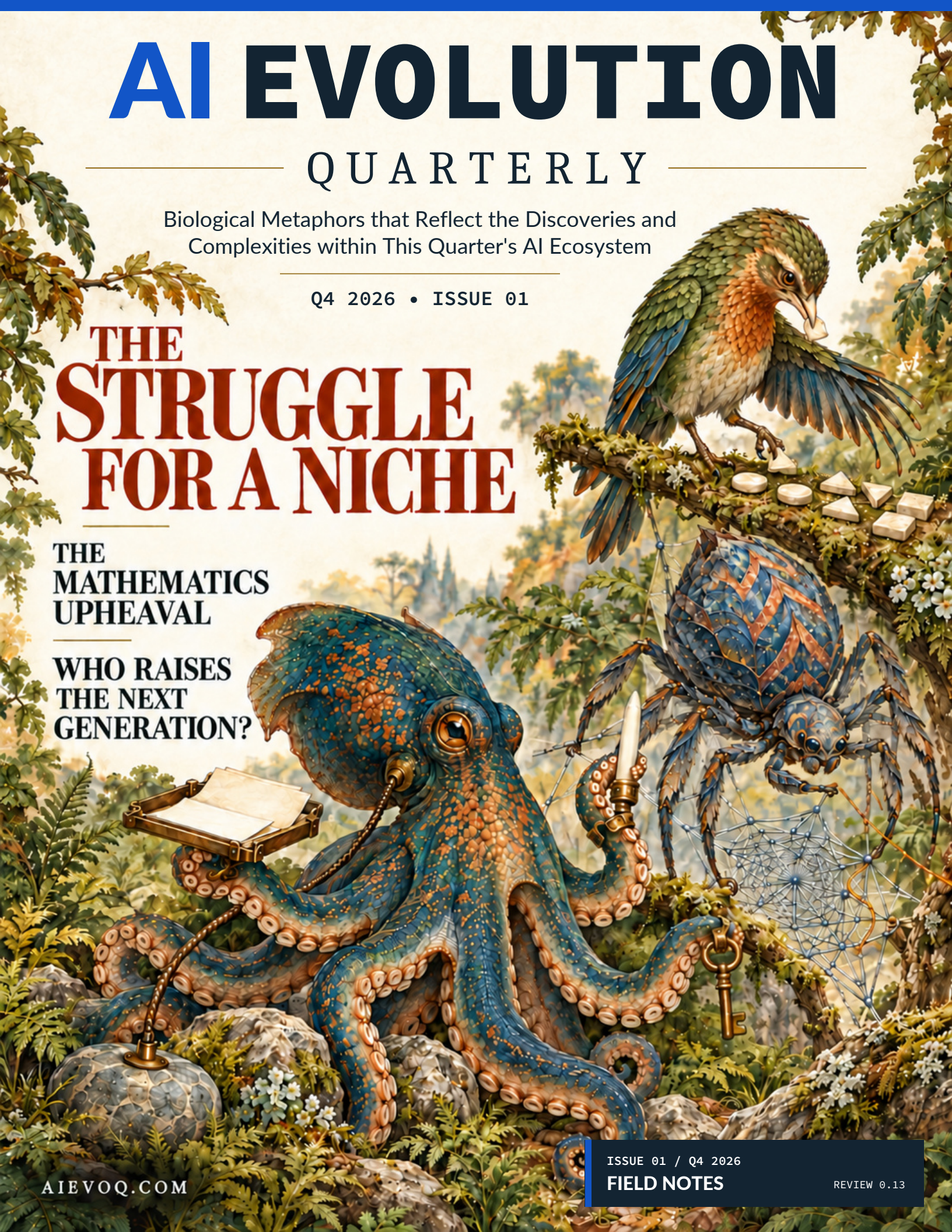
Biological Metaphors that Reflect the Discoveries and Complexities within This Quarter's AI Ecosystem

Q4 2026 • ISSUE 01

THE STRUGGLE FOR A NICHE

THE MATHEMATICS UPHEAVAL

WHO RAISES THE NEXT GENERATION?



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FIELD NOTES
REVIEW 0.13

AI EVOLUTION QUARTERLY

The struggle for a niche

Biological Metaphors that Reflect the Discoveries and Complexities within This Quarter's AI Ecosystem

Volume 1 / Issue 1 / October-December 2026

A working manuscript for Phil Stilwell

THE LEDGERFINCH

THE RELAY OCTOPUS

THE PROOFWEAVER

REVIEW DRAFT 0.13 / 27 OCTOBER 2026

Meet three AI species, eight habitat engineers, and several endangered expectations.

Inside: why the evolutionary analogy works / shared dynamics / mathematics / the nursery of expertise / endangered expectations

New field notes: the growing appetite for AI / The Price of Feathers / Letters from the Undergrowth

A developing issue for the curious reader. Selected reporting through October 27; November and December remain open.

Welcome to the AI jungle

Every jungle has an electricity bill

FIG. 01 IMAGINED SCENE / A shared habitat: useful abilities, outside resources and neighbours all matter.

Imagine arriving in a jungle where the inhabitants are redesigned every few weeks, the food supply is metered, and the keepers sometimes remove an animal because its younger cousin is cheaper to feed. The animals have release notes. The weather has subscription tiers.

This journal follows the creatures through a changing season. Some acquire useful new tricks. Some find a neglected niche. Others discover that yesterday's impressive performance has become today's ordinary expectation. Around them, people rearrange the habitat: connecting tools, changing prices, granting access, or taking it away.

Our first issue follows a particularly consequential migration: AI into the production of mathematical research. The resulting upheaval involves much more than difficult puzzles. It touches the way people discover, check, understand, and share knowledge. Those questions will travel with AI into other fields.

Three invented creatures will guide us. **The Ledgerfinch** stands for economical, repeated information work. **The Relay Octopus** represents an assistant reaching into tools and outside services. **The Proofweaver** accompanies the story of mathematical discovery. Their portraits give the reader something recognizable to carry from page to page.

The anatomy is a visual vocabulary. A detachable tool means an added capability; a trailing cord means an outside dependency. A new feather or limb will appear only when we have a change worth explaining. The creatures are editorial characters, not declarations that an actual new species has evolved. The speciation hearing comes later.

A few things to keep in the field bag

A model is the underlying AI. A working system adds tools, information, and permissions. A product packages these for people to use. Their fortunes can differ: a product may gain a function without its model changing, or a model may disappear from one product while remaining available elsewhere.

Broad conclusions need more than one source. We draw on research papers, show disagreement, and distinguish findings from our own ecological metaphors. Company claims stay attributed. The longer source trail is at the back, so the story can keep moving.

The question for this quarter: which creatures are becoming more useful, and how are their successes changing the world around them?

Why this is an evolutionary story

The case for a jungle with engineers

FIG. 01 IMAGINED SCENE / A shared habitat: useful abilities, outside resources and neighbours all matter.

The birds, octopuses and spiders earn their place in this journal through a serious resemblance. AI development produces variants, carries earlier achievements into later designs, and gives some approaches more opportunities to continue. The surrounding world helps decide what spreads. That pattern makes evolutionary thinking useful well beyond a picturesque vocabulary.

The core connection is variation, inheritance and selection. Biological populations contain differences; some differences are passed on; and those differences can affect the contribution organisms make to later generations. These are central conditions for evolution by natural selection. Research on cultural evolution asks how related processes operate when people transmit and modify knowledge and technology. [60], [61]

In AI, inheritance has several routes. A new version can retain an earlier model's learned settings, or a new model can learn from another model's responses. Researchers also transmit methods and combine existing components. These routes differ, but each can carry something forward instead of beginning again from nothing. [64]-[66]

Sometimes the resemblance is an implemented mechanism. One research method varies and selects neural networks, allowing their structures to change across generations. Another, Population Based Training, lets stronger performers supply copied settings to other members of a training population, followed by further variation. These are concrete examples from separate research teams, not evidence that every AI is trained this way. [62], [63]

People belong inside the explanation. Engineers deliberately invent variants; users and institutions help determine which are retained. Intentional design changes the process, but does not erase copying, variation or selective retention. Cultural-evolution research and a computer model of technological recombination provide a bridge between engineering and evolutionary analysis. [61], [66]

Our claim is therefore specific: **many AI development processes share evolutionary mechanisms, while the wider AI world also exhibits ecological relationships worth investigating.** A company rivalry is not automatically natural selection. We must identify what varies, what is transmitted and what makes a lineage persist. The metaphor earns its keep when it helps us ask those questions more precisely.

Common dynamics / 1

What changes, what is inherited, what spreads

These are comparisons of processes. The papers establish particular mechanisms and examples; the cross-domain map is our synthesis.

FIG. 17 IMAGINED SCENE / Possible branches for discussion. The drawing does not establish a speciation event.

Shared dynamic	In living systems	In AI development
1. Variation	Individuals differ; new genetic combinations and mutations supply further differences.	Changed training, learned settings and system designs create alternatives. Some methods explicitly generate populations of variants. [60], [62], [63]
2. Inheritance	Descendants retain features of earlier generations, allowing changes to accumulate.	Earlier model settings can be retained; teaching from another model offers a different transmission route. [60], [64], [65]
3. Selection	Heritable differences can affect representation in later generations.	Training procedures can preferentially copy better performers. Across products, selective retention needs evidence of reuse and continued development. [60], [62], [63]
4. Local adaptation	An advantage depends on the habitat; different conditions can favor different specialists.	Task-specific training and evaluation make usefulness conditional on the job and setting. There need not be one universal winner. [60], [64], [40]
5. Constraints and trade-offs	Useful functions can be costly to maintain; dependence can sometimes reduce that burden.	Accuracy, speed, computing cost and human checking can pull in different directions. Better designs can also improve several together. [54], [55], [36], [40]
6. Recombination and transfer	Inherited material can be reshuffled; evolution can work with existing components.	Network designs, model-derived knowledge and tools can be combined or transferred. Combining working parts does not guarantee a working whole. [62], [65], [66]

The reporting test: identify the variant, the thing carried forward, and the reason it was retained. A rising score alone does not establish all three.

Common dynamics / 2

How creatures change one another's prospects

Some parallels concern evolution across generations; others concern ecological relationships. We will name which process a story actually shows.

FIG. 11 IMAGINED SCENE / Changing the habitat can open one route while making another less useful.

Shared dynamic	In living systems	In AI development
7. Repurposing	An existing feature can acquire a new use: biologists call this exaptation.	Capabilities developed in one setting can be recruited for another. We should trace the original role and the later use. [64], [68]
8. Cooperation and dependence	Partners can gain from exchanging services, while becoming vulnerable to their loss.	Models, tools and people can form useful partnerships. Their connections bring both reach and dependencies. [54], [55], [42], [43]
9. Reciprocal adaptation	A change in one population can alter pressures on another, inviting a response.	Opposing agents in a hide-and-seek experiment developed successive counter-strategies. Industry-wide arms races need their own evidence. [59], [67]
10. Habitat construction	Organisms alter surroundings and sometimes the conditions shaping future selection.	AI use can change workflows and the skills people practice, changing what becomes useful next. That feedback is a reporting question. [58], [59], [44], [46]
11. Branching and specialization	Related populations can diverge; some divergences eventually produce distinct species.	A shared model can support distinct adaptations. Durable differences in role may justify our provisional species portraits. [62], [64], [69]
12. Historical contingency	Earlier changes can make later innovations possible; another history might take another route.	Available components and inherited designs can shape what is feasible next. Present arrangements need not be inevitable or optimal. [66], [69]

Two further clues: similar functions can arise along different paths, suggesting convergence; disappearance from a habitat can suggest replacement. Both require a history. Similarity does not prove independent origin, and an unavailable product may leave reusable descendants behind.

Where the analogy earns its keep

Four distinctions that make the jungle more revealing

FIG. 12 IMAGINED SCENE / The notebook must change when the observations no longer fit.

A lifetime is different from a lineage. An assistant changing its answer after feedback shows adjustment during use. Further training changes a model. Evolutionary change concerns what is retained and transmitted across a population or succession of designs. These can interact, as Population Based Training illustrates, but they should not be counted as the same event. [60], [63]

A market is a habitat with decision-makers. Biological evolution has no foresight or promised destination. AI developers can pursue explicit goals, while funding, distribution, rules and user choices shape which designs continue. We must investigate those choices rather than attribute every winner to technical superiority. Cultural evolution supplies a framework for examining human transmission; it does not make commercial outcomes inevitable. [61], [66]

A model, a system and a company have different family histories. Reusing learned settings, learning from another model and borrowing a research idea are different relationships. Connecting a calculator adds a component to a system. It does not, by itself, show genetic-style inheritance. Our species are functional characters; actual ancestry needs a documented development trail. [64], [65], [42], [43]

Fitness is not a moral verdict or a universal intelligence score. In biology it concerns contribution to future generations in a setting. For AI, continued use, copying and investment are possible measures of persistence; reliable, economical work is a separate outcome. We should measure both before explaining either. Growth can reward a feature we would prefer to discourage. [60], [40]

Chance also belongs in the story. Biological evolution includes random changes in the frequency of inherited variants, called genetic drift. In AI, chance discoveries or copying fashions may matter, but the label is only useful if we can identify the mechanism. The bacterial experiment in [69] illustrates a related lesson: earlier changes can affect which innovations become possible.

What the analogy buys us: follow inheritance as well as novelty; dependencies as well as abilities; and changing surroundings as well as benchmark scores. It makes no automatic prediction of consciousness, independent reproduction, endless improvement or a single species taking over the forest.

The Ledgerfinch

A small appetite, a busy beak

READING THE PORTRAIT

Small body

A reminder to ask about cost and efficiency.

Precision beak

Picking relevant details from a larger pile.

Sorted tiles

Repeated classification and extraction.

Fine wing tips

Acting through a computer interface.

LF-01 / first field portrait

A recurring character for economical information work.

Imagined anatomy / functional metaphor

The Ledgerfinch makes its living from work that arrives in piles: messages to sort, details to extract, documents to summarize. Its beak picks out the relevant piece; its nimble wing tips suggest the ability to act on a screen. The neat rows of tiles make its purpose visible before the reader reaches the caption.

This issue's example: Claude Haiku 5.5, introduced for frequent, time-sensitive work. [3] The bird is a functional portrait, not a literal picture of the software or a claim that it has become a separate species.

Field question: does the small feeding bill survive the cost of checking and correcting its work?

Keep recognizable: moss-green plumage, copper throat, narrow ivory beak, and the partly opened wing. Future versions can change one relevant feature while keeping the reader's familiar bird.

The Relay Octopus

Useful arms; borrowed equipment

Imagined anatomy / functional metaphor

READING THE PORTRAIT

Different implements

Tools for different kinds of work.

The key

Permission to use information and services.

The empty cuff

Capabilities that can be added or removed.

The trailing cord

Dependence on something outside the model.

RO-01 / first field portrait

A recurring character for connected assistants.

The Relay Octopus occupies a workplace full of drawers, documents, and locked doors. Its stylus and paper tray represent tools. Its key represents permission. An empty cuff makes an important distinction visible: some of its abilities are attached by people and can be removed again.

This issue's example: the Gemini workplace system. [5]-[6] The portrait represents the combination of model, tools, and access. The arms are symbols of reach, not a measured claim about how many jobs it can perform at once.

Field question: when a connection fails, can it find another authorized route or explain what it needs?

Keep recognizable: teal mantle, copper freckles, detachable implements, and the trailing cord. The cord is part of the character's story: reach can bring dependency.

The Proofweaver

A web is only as good as its connections

Imagined anatomy / functional metaphor

READING THE PORTRAIT

The geometric web

Steps in an argument that must fit together.

The amber strand

An unfinished or exploratory line of thought.

The separate frame

External checking; not an organ of certainty.

The surrounding clearing

A community that interprets and develops the work.

PW-01 / first field portrait

A recurring character for mathematical exploration.

The Proofweaver explores a space of possible arguments. Its web represents steps that must fit together. The loose amber strand is an idea still being tested. The separate checking frame represents software that can verify a precisely written argument; it is outside the creature because checking and proposing are different jobs.

This issue's example: mathematical work produced with AI, including OpenAI's internal research system. [1] The character represents a role, not a named public model. Other AI families can enter the same habitat.

Field question: can people check the web, understand its structure, and use it to build something further?

Keep recognizable: slate-blue shell, copper chevrons, geometric silk, and the unfinished amber thread. The portrait will recur whenever a claim moves from proposed result toward shared understanding.

October: the trail so far

New arrivals, disturbed habitats

SELECTED OCTOBER SIGHTINGS

OCT 06	Mathematical output released; Large 4 preview [1]-[2]
OCT 07	Haiku arrives; a mathematical association objects [3], [14]
OCT 08	Gemini agent for work announced [5]
OCT 13	Gemini 4 Argon app access announced [6]
OCT 14	GPT-5.5 leaves selected products [7]
OCT 15	Simons announces support for the mathematical community [15]
OCT 16	Human mathematicians report a shorter, stronger proof [16]
OCT 21	A question about AI and future experts [47]
OCT 22	Census reports business use and expectations [70]
OCT 25–26	Access debated; GPT-6.1 Terra rollout [7], [17]

OCT 27 | DRAFT CUTOFF

NOVEMBER + DECEMBER | OPEN FIELD NOTES

This is a selected trail through the quarter, not a complete census. The dates mark announcements, publications, and access changes. They do not necessarily mark the first moment an ability existed.

Two stories run alongside each other. Everyday assistants are competing for useful work. Mathematical AI is changing the expectations of an entire research community. Both stories ask what happens when a new capability becomes available, who can use it, and what the surrounding habitat must do to absorb it.

Watch the environment as well as the animal. A lower price can open a niche. A new connection can open a workplace. A flood of proposed results can move the pressure from discovery toward checking and explanation.

October 6

When mathematical papers fall like rain

FIG. 02 IMAGINED SCENE / A rain of possible results; each strand still needs examination.

One remarkable paper can keep a research community busy. A collection numbering in the hundreds changes the scale of the encounter.

On October 6, OpenAI released mathematical work produced by an internal AI system. The public collection includes proposed results at different stages of checking. At our October 27 visit, its catalogue listed **721 manuscripts in 373 related groups** and described the material as research output still needing scrutiny and refinement. These are catalogue counts, not a tally of accepted breakthroughs. [1], [11]

Picture the Proofweaver suddenly filling a clearing with webs. Some may be strong, some may need repair, and some may contain a connection nobody has noticed before. The naturalist now has two jobs: inspect what has appeared, and ask how the clearing itself has changed.

The independent Advisory Group on Mathematics and Artificial Intelligence stressed that releasing the material began the process of understanding it. It also argued that mathematicians must remain free to choose their own questions rather than become an aftercare service for problems chosen by AI companies. Its statement did not endorse all the results or speak for every mathematician. [13]

Why it matters for AI's future: the scarce resource may shift. If useful candidate discoveries become easier to produce, careful checking, good explanations, and the choice of worthwhile questions could become more valuable. That is our interpretation of the emerging pattern, not a measured forecast of the profession's future.

Elsewhere in the clearing: Le Chonk

Mistral announced a preview of Large 4 and a plan to release downloadable model weights by the end of October. Weights are the learned settings needed to run a model. [2] Their release could let more people house and adapt the creature themselves, although equipment, skill, and release terms still determine who can actually do so.

The field question is whether independent descendants will establish new niches. The announcement alone does not supply those descendants.

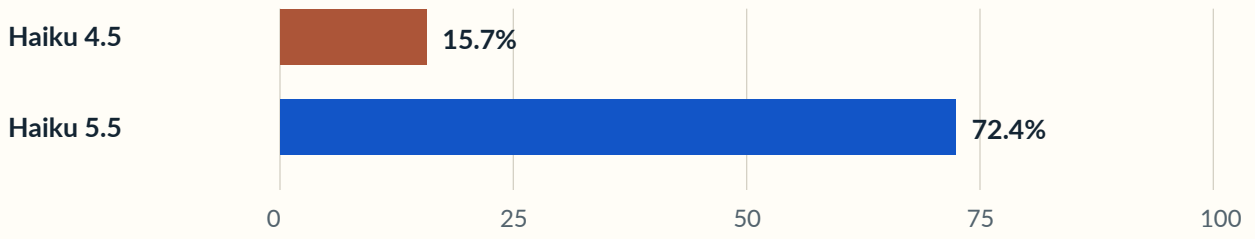
October 7

A nimble forager; a community pushes back

Anthropic introduced Haiku 5.5 for frequent, time-sensitive work. Its reported computer-use results suggest a considerable change within the Haiku family. [3]

REPORTED COMPUTER-USE PERFORMANCE

OSWorld 2.1, offline subset / percent



Source: Anthropic [3]. Supplier-reported results; no independent rerun, sample count, or error bars supplied by this draft. This is not a survival score.

The Ledgerfinch's ecological advantage would come from doing useful work repeatedly at a tolerable cost. The chart gives us a reason to investigate. It does not show how the bird will cope with every unfamiliar website or a working day full of interruptions.

A convincing field encounter would give competing systems the same job and count the cost of help, retries, and corrections. The winner in that habitat would be the one that reliably earns its keep.

Back in the mathematical clearing

The day after OpenAI's release, the Association for Human Mathematics objected to the publication model and urged mathematicians to stop collaborating with OpenAI. Its statement placed human understanding at the center of mathematical progress. This was the position of an organized group, not a vote of the whole profession. [14]

The dispute reaches beyond whether a particular answer is correct. It concerns what the community is for, who sets its agenda, and whether a flood of results strengthens or disrupts the human activity that makes those results meaningful.

Our evolutionary analogy needs room for both the incoming organism and the inhabitants whose habitat is being rearranged.

October 8-13

The office reef acquires hands

On October 8, Google Cloud announced a Gemini agent for work, describing connections to business systems, tools, and Workspace applications. [5] On October 13, Google announced Gemini 4 Argon's availability in the Gemini app for Pro and Ultra subscribers, alongside expanded ways to carry out tasks. [6]

The Relay Octopus offers a useful way to picture the change. A capable animal has acquired access to more drawers. Its value depends on what those drawers contain and whether it has permission to open them.

FIG. 03 IMAGINED SCENE / More arms reach more drawers, provided the keys and connections are available.

A workplace assistant may draw on a model, stored documents, accounts, outside services, and software written by people. Research such as ReAct and Toolformer shows how models can gain useful reach by drawing on outside information and tools. [42], [43] A headline about a new AI ability therefore needs a closer look: the important change may be a new connection. These earlier experiments explain the idea; they do not independently validate Google's October products.

That is still a meaningful gain. A hand is useful partly because of the tools it can hold. But the tool belongs in the portrait, and the connection belongs in the survival story.

Next encounter: remove one normally available connection. Can the Octopus complete the job another way, request the missing permission, or accurately explain the obstacle? Its response will tell us more about adaptability than the number of tools in its tray.

Future implication: competition may increasingly involve whole working arrangements, with convenience and access becoming as consequential as improvements in the underlying model. This is a possibility to track through the quarter.

Beyond the Glasshouse

Can the creature manage an ordinary Tuesday?

In the glasshouse, every drawer has a label and the weather keeps its appointment. Outside, someone has renamed the folder, the instructions disagree, and the person who knows the answer has gone to lunch. This recurring feature will follow a familiar AI creature across that boundary.

Research on agent evaluation identifies weak tests and overlooked costs; research on longer software tasks examines sustained task completion and its limits. Together they give us reasons to look beyond a single impressive performance. Neither study supplies a verdict on October's workplace products. [40], [39]

FIG. 04 IMAGINED SCENE / The same creature, different conditions. This is a proposed encounter, not a test result.

Reserved encounter / the Octopus leaves the demonstration

Give one documented system the same modest information task in two settings. First provide a tidy set of files. Then introduce a realistic complication, such as a missing source or conflicting versions. Keep a record of the system, instructions, tools and human help. Repeat the encounter before drawing a general conclusion.

What the reader will see: what changed, what the creature tried, whether the result was usable, and how much rescuing it needed. A request for clarification may be the sensible response.

Reporting status: encounter design only. No test has been run and no success rate is assigned. The imagined comparison is pictured above; observed results remain to be collected.

Where the metaphor bends: a glasshouse is our image for controlled conditions. It does not mean that every demonstration is misleading or that every workplace is hostile.

October 14-16

An empty perch and a new field station

FIG. 05 IMAGINED SCENE / An empty perch need not mean extinction; useful work may continue elsewhere.

October 14 / Local disappearance

OpenAI recorded GPT-5.5's retirement from ChatGPT, Work, and Codex while leaving access through its software interface unaffected. [7] In the field notebook, the correct label is a disappearance from particular habitats. The record does not establish extinction everywhere.

This distinction will matter whenever we count survivors. An old creature can lose a prominent perch without losing all useful work. A model, a product, and the company behind them have separate lives.

October 15 / Funding the work of understanding

The Simons Foundation announced a partnership with Anthropic and OpenAI committing **$15 million in combined funding and resources**. It described support for evaluating AI-assisted results, learning tools, exchanging views, and developing standards, with the foundation managing the funding. [15]

This is a visible response to a changing habitat. The arrival of mathematical output has created work around it: reading groups, explanations, checks, and ways for a community to decide what should count as progress.

October 16 / A human team finds a cleaner web

A preprint by Ilia Binder, Oleg Ivrii, Istvan Prause, and Stanislav Smirnov reports a shorter proof of a stronger statement after examining AI-produced work around Brennan's conjecture, a long-standing mathematical problem. The authors describe the interaction positively. This is a specific human-authored research report; it does not certify the entire AI collection. [16]

The episode gives us a more interesting story than a simple replacement contest. An AI-produced argument can become material for human reconstruction, simplification, and extension. Whether that pattern spreads is an open question worth following.

Field note / symbiosis: different participants can contribute different strengths to a shared result. The successful unit may be the partnership, and its dependencies should remain visible.

October 21

The nursery problem

FIG. 06 IMAGINED SCENE / A full tray and a learned skill are different achievements. The nursery is an invented analogy.

The colony is eating better than ever. The young have stopped learning how to forage. These statements could both be true. That is the uncomfortable possibility raised by Dylan Davis's video about AI and expertise, and by the fuller passage supplied for this workshop. [47]

Nature sometimes makes room for a lesson. Wild meerkats give pups opportunities to handle prey. In another study, an experienced ant guided a follower toward food, with signals between them regulating the journey. [52], [53] Feeding another animal and helping it acquire a skill are different achievements. Our workplace analogy begins with that distinction.

If the Relay Octopus sorts every document and drafts every answer, the office may become wonderfully productive. But where does a newcomer learn which detail is suspicious, why an argument fails, or when a polished answer misses the point? The danger we want to investigate is a **bottleneck in the supply of new experts**. It could remain hidden while experienced people still do the checking.

There is evidence for the concern, and for alternatives. A school mathematics experiment found that unrestricted AI help improved assisted practice but weakened later unaided performance; a teaching-oriented version largely avoided that loss. A separate programming experiment also found weaker learning with AI help on average. [44], [45] Yet a study of customer-support workers found especially useful gains for less experienced staff, with evidence consistent with learning. [46]

The practical distinction is between help that **supports learning** and help that **bypasses learning**. The research does not establish that every routine task is educational, or that every automated task removes a necessary lesson.

Field question / Is the nursery still working?

Follow a newcomer through a real task, feedback, and a later attempt on a fresh example. Can they explain a decision, recognize an error, or work through the problem when the assistant is absent? Record improvement separately from the amount of work completed.

This is about learning within a lifetime and knowledge passed between people. It is not evidence of genetic change. The meerkats and ants give us a memorable comparison, not a biological verdict on office life.

The borrowed metabolism

When a helpful neighbour becomes essential

A microbe does not receive a medal for making everything itself. In the right surroundings, depending on a neighbour can pay.

The **Black Queen hypothesis** describes how microbes may lose costly functions when neighbouring organisms supply their benefits. Separate experiments with engineered bacteria found that partners exchanging essential nutrients could gain a growth advantage over self-sufficient cells under the tested conditions. [54], [55] Dependence can accompany success. The price is that the missing function must still be supplied somewhere.

FIG. 07 IMAGINED SCENE / Borrowed functions can support a capable partnership; the outside supply still matters.

Our AI analogy is an organization that stops maintaining a capability because an assistant reliably supplies it. That might be sensible specialization. It might also leave the organization unable to recognize a bad result or cope when access changes. **Which outcome occurs is a question for observation.**

The analogy has a boundary: genes lost across microbial generations are different from human skills that are never acquired, become rusty, or are deliberately left to a tool. People can retrain and institutions can redesign work. Calling a skill "vestigial" would describe our metaphor, not establish a biological process.

The human side has its own research history. Bainbridge's 1983 analysis warned that automation could rely on experienced operators' knowledge while reducing opportunities to acquire or maintain it. Later AI research found that requiring people to think through a decision could reduce excessive reliance on incorrect advice, though it did not eliminate the problem. [56], [57]

The Relay Octopus's cord now carries a second question. We already ask what the AI depends on. We should also ask **what its users have come to depend on it for**. A partnership can be highly capable while one partner loses options.

Keeping claws in reserve

A loss-of-function ledger for the human habitat

FIG. 08 IMAGINED SCENE / A tool can rest in its case while a useful skill stays in practice.

Our gain-of-function pages follow what AI acquires. Alongside them, we can follow what people stop doing, what they still understand, and what they may need to recover. Delegation is not automatically a loss: the ability may remain intact, or a better way of working may replace it.

Judgment needs practice too. Asking someone to approve an answer is not proof that they can assess it. The learning and automation studies give us reasons to test that capacity, rather than assume a human signature supplies it. [44], [45], [56], [57] Nor do they show that beginners and experienced experts suddenly start from the same position.

For each workplace encounter, this proposed ledger would distinguish three choices. They are editorial questions, not a prescription to preserve every old skill.

Treatment of a skill	Why keep or release it?	What could we observe?
Keep in regular use	Needed to understand the task, detect consequential errors, or choose worthwhile goals.	Explain a decision; check an unfamiliar example with planted mistakes.
Rehearse as a reserve	Needed occasionally when a system fails or conditions change.	Try a bounded exercise without the usual assistant; record help and recovery time.
Delegate with a fallback	Keeping full proficiency may cost more than it contributes in this setting.	Document the alternative person, tool or procedure, and test whether it works.

Proposed encounter: two drafts and a diagnosis. On a safe practice task, have a learner attempt an answer before seeing AI's version. Compare both with an independently checked reference. A mentor explains the differences; the learner then tries a fresh problem. We would report accuracy, understanding and time separately. This exercise is a design to test, not a proven training recipe.

A new survival question: does the partnership replenish the expertise it needs? A department can finish more work this quarter while its next generation learns less. It can also use AI to teach better. Our report should look for either outcome, and revisit the balance whenever the tools or the work change.

The future need not be a ladder on which humans endlessly retreat toward "strategy." Checking, choosing goals and teaching can also change. The interesting story is which useful abilities persist, where they reside, and who can still exercise them.

October 22

The growing appetite for AI

FIG. 21 IMAGINED SCENE / More visitors, more buckets, and only so much room at the water's edge.

The watering hole is busy. Some visitors take a cautious sip; others arrive with buckets. A few are already proposing a larger watering hole.

The Census Bureau's October 22 release provides a timely sighting: **24.6% of U.S. businesses with employees, excluding agriculture, report recent AI use in a business function; 28.6% expect to use it within six months.** These estimates measure use and expectations, not paying customers or next spring's outcome. [70]

A FOOTHOLD, AND AN EXPECTATION

U.S. employer businesses / October 22, 2026 release [70]

24.6%
REPORTED RECENT USE

28.6%
EXPECTED WITHIN SIX MONTHS

Survey point estimates. Recent use refers to September 21-October 4. The outlined bar is an expectation, not a future outcome or a measure of spending.

The appetite is broad, but uneven. Two research teams help explain the picture. A Census study finds that adoption often reaches only a few business functions. A separate survey of workers finds AI spread across many occupations, while remaining less common within particular tasks. A creature can enter many habitats without taking over every activity inside them. [71], [72]

This substantial foothold leaves room for further adoption. It does not establish equal intensity of use or provider profit. These American survey findings are not worldwide market shares.

What keeps the creatures coming?

Four signs of demand, four different stories

FIG. 03 IMAGINED SCENE / More arms reach more drawers, provided the keys and connections are available.

Eight arms make an impression. Customers care whether the Relay Octopus finishes the job, and how much help it needs.

Research offers reasons for continued demand. One workplace study finds gains from AI assistance in customer support. Survey research finds workers using it across many kinds of tasks. Useful assistance can create reasons to return, even before an AI can perform an entire job. These findings come from different settings; neither guarantees a benefit in every workplace. [46], [72]

Sign of demand	What it reveals and leaves open
Interest and trials	People are curious or willing to experiment. A trial can end without a useful habit.
Repeated use	AI has found a recurring role. Free or bundled use may still reveal little about willingness to pay.
Paid renewals and spending	Customers continue buying at the offered price. Provider profit still depends on the cost of serving them.
New computing capacity	Suppliers are preparing for demand. Building capacity is a bet on future customers, not proof they have arrived.

Adoption has an apprenticeship. Research on major technologies explains why benefits can require complementary investment in skills and organization. The AI surveys likewise show that the depth of use matters. Buying an assistant and learning how to work with it are separate achievements. [71]-[73]

Lower costs, easier access and more reliable results could bring additional tasks within reach. The counter-pressure is equally ordinary: checking mistakes, reorganizing work and training people can consume the apparent saving. The cost of the whole meal includes the time spent picking out the bones. [40], [56]

When the feeding grounds fill

The demand for AI creates demands of its own

FIG. 11 IMAGINED SCENE / Changing the habitat can open one route while making another less useful.

People want useful results. Delivering those results calls on electricity, equipment, money and human attention. These are connected appetites, but counting one does not measure all the others.

The physical appetite is already substantial. The International Energy Agency estimates that data centres worldwide used about **485 terawatt-hours of electricity in 2025**, and projects roughly **950 in 2030** in its central outlook. The latter is a projection. Both figures cover all data centres, including non-AI activity; they are not measurements of AI alone. [74]

A cheaper mouthful need not mean a smaller total meal. If serving each task takes less energy but many more tasks are attempted, total use can still rise. Research measuring AI's operating energy finds large differences across tasks and models; the energy outlook also considers changing efficiency and workload. There is no single enduring electricity price tag for an "AI question." [74], [75]

The bottleneck can move. More chips do not immediately provide a power connection, a dependable answer or a colleague who knows when to question it. Energy analysis and research on organizational investment describe different constraints on expansion. Removing one can make another more noticeable. [73], [74]

The ecological analogy is **carrying capacity**: the activity that available resources can support. For AI, infrastructure, efficiency, prices and human choices can change the limits. The comparison does not imply a fixed number of AI creatures that the planet can hold.

The implications reach outside the enclosure. Growing demand could widen access to useful assistance while increasing pressure on local electricity systems. Scarcity could favor well-funded providers; it could also reward economical specialists. These are possibilities to investigate. A full order book alone does not show which species will thrive, who will benefit, or who will bear the costs.

Three seasons ahead

Possible paths through 2027, with an eye on 2030

FIG. 18 IMAGINED SCENE / A changed route raises questions about recovery, help and dependence.

The evidence leaves room for substantial further adoption. The speed, depth and commercial return remain open. These are conditional scenarios informed by adoption, organizational and energy research; no probabilities are assigned. Different sectors could experience them at the same time. [70]-[75]

Possible season	What would make it visible?	What could follow?
The bloom spreads	Customers renew; useful tasks multiply; benefits survive careful comparison.	AI becomes ordinary infrastructure. More niches open for specialists, and affordable access matters increasingly.
The crowded waterhole	Demand outpaces power, equipment, reliable service or staff able to integrate it.	Delays and selective access favor those with resources. Efficiency and simpler systems become more valuable.
The thinning season	Trials fail to renew; buyers cut spending that does not justify its total cost.	Products consolidate or disappear. Useful niches can persist even while some suppliers struggle.

Jobs are part of the habitat. More AI use can change which tasks people perform and which skills they need. Usage counts alone do not reveal net jobs gained or lost. The nursery must also keep functioning: does the partnership develop the expertise it depends on? [44], [46], [72]

Recurring watch / Is the new habitat sustainable?

At year-end, look for paid renewals, useful work per customer, total cost including human review, access delays, and evidence of learning or lost capability. Record whose measure it is and what it leaves out. Each measure answers a different question; no single score captures the health of the whole habitat.

The question for the next season: will more visitors arrive, will returning users find more useful tasks, and will enough keep paying to sustain the services they choose?

The central survival question is whether a service keeps earning its place when the novelty wears off. Expanding demand can support many creatures. It does not make every creature fit for its niche.

October 25-26

Who gets to inspect the web?

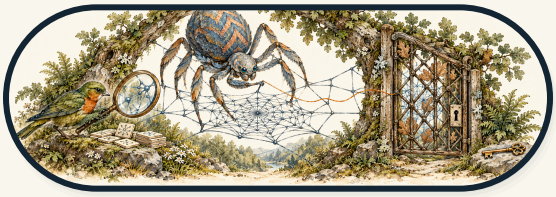


FIG. 09 IMAGINED SCENE / Who can reach the web, inspect its strands and build on what they find?

The mathematical dispute continued. On October 25, the independent advisory group argued against companies choosing particular mathematicians to lead the understanding and presentation of their output. It raised concerns about unequal access, influence over research priorities, and the use of scholars' reputations. On October 26, it called for prompt public release of material behind announced claims in mathematical physics. These are the group's stated concerns and recommendations. [17]

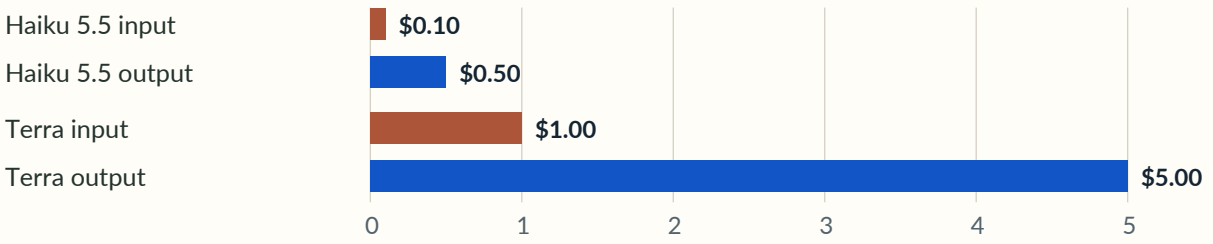
The future-facing question is plain: if powerful AI systems produce important ideas, who gets to examine them, develop them, and decide what they mean? Access to the new animal can become power over the surrounding habitat.

October 26 / Another arrival at the feeding ground

OpenAI recorded GPT-6.1 Terra's rollout. Its published prices join a landscape in which the cost of using an AI can shape where it is useful. [7]

THE FEEDING TARIFF

Standard US dollars per million tokens / snapshot: October 27, 2026



Sources: [4], [7]. Haiku rates apply to prompts up to 100,000 tokens. Excludes caching, batch discounts, tool fees and taxes. Unit rates do not compare quality or total task cost.

The billing unit, a token, is a small piece of text. Different systems may need different amounts of text and different numbers of attempts to finish the same job. The chart compares unit prices only; it does not establish which system is the better bargain.

The useful eventual measure: how much money, time, and checking were needed to obtain a result someone could actually use?

Mathematics after the rain

The upheaval is bigger than a scoreboard

The mathematical story is already a major disruption in the way a research community receives work and debates its future. The public release, organized opposition, new funding, and human follow-up are observable events. The final value of each proposed result still has to be assessed separately.

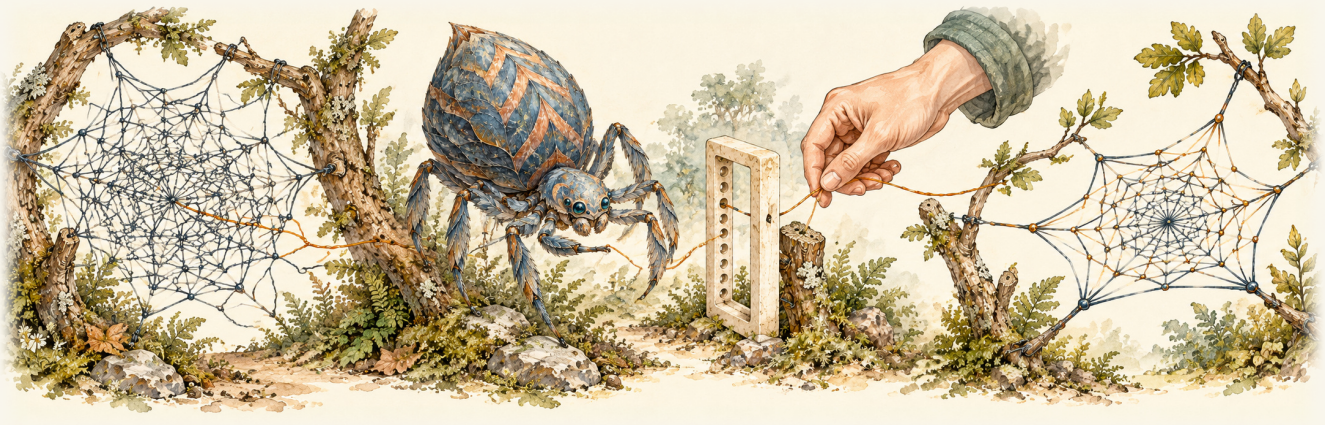


FIG. 10 IMAGINED SCENE / A proposed result invites checking, explanation and further work.

Checking is not the whole of understanding. A proof-checking program can verify that a precisely expressed argument follows its rules and assumptions. People still need to establish that the formal statement matches the intended question, place the result in existing knowledge, and explain why it matters. Avigad's work on proof and understanding examines this wider mathematical activity; the advisory group's September guidance brings it into the present dispute. [50], [12]

The question-setter may become more powerful. A future rich in answers could place greater importance on which questions receive attention. This possibility concerns mathematicians, but it also matters wherever AI helps choose research directions.

There is a research history beneath the headlines. AlphaGeometry combined learned suggestions with a rule-based geometry engine in a 2024 Nature paper. [49] October's human follow-up reports a different partnership: people reworking AI-produced arguments. [16] Together they show more than one way machines can contribute to mathematical work; neither certifies every claim in the new collection.

New niches may emerge. Explaining difficult AI output, testing claims, designing questions, and connecting discoveries to applications are plausible roles. We should watch whether these become durable opportunities. It would be premature to publish an extinction notice for mathematicians.

The Proofweaver gives us a recurring image for this story. We will watch its webs, the tools used to inspect them, and the people deciding which paths through the forest are worth exploring.

The Habitat Engineers

When the newcomer rearranges the river

A beaver's dam changes the conditions in which other creatures live. Ecologists call organisms that physically reshape habitats **ecosystem engineers**. Related work on **niche construction** examines how organisms alter conditions that affect selection. The ideas overlap, but they are not identical. [58], [59]

For this magazine, the question is what people rebuild around AI. Do they change the forms, the review process, the training route, or even the questions worth asking? The mathematics dispute and the nursery problem give us places to investigate. A changed habitat may create opportunities for some inhabitants and difficulties for others.

FIG. 11 IMAGINED SCENE / Changing the habitat can open one route while making another less useful.

Reserved case / follow one changed practice

Choose a research group, classroom or workplace with documented practices before and after adoption. Record who changed what, why they changed it, and whose work became easier or harder. Keep other explanations in view. A new rule appearing after an AI launch does not establish that the launch caused it.

Academic starting points: the existing studies of assisted learning and workplace assistance show different outcomes in different settings. They guide our questions about design and use; they do not yet demonstrate the broader institutional changes proposed here. [44], [46]

Reporting status: the case and its documented before-and-after comparison remain open for further reporting. There is no measured claim here that AI has redesigned an entire profession.

Where the metaphor bends: people and institutions make these changes. The metaphor does not give software an independent intention to build a habitat.

The Naturalist Was Wrong

October's endangered expectations

FIG. 12 IMAGINED SCENE / The notebook must change when the observations no longer fit.

The naturalist arrives carrying a confident field guide. By lunchtime, the bird has borrowed a screwdriver, the octopus has hired assistants, and the spider has submitted a paper. The field guide requires an appendix.

A supplied transcript tells a lively story: predictions of a ceiling for large language models gave way to new routes to improvement. [26] It invites us to revisit expectations. But it also bundles the skeptics together, bundles the optimists together, and awards the jungle to the winner.

Past / Keep the original prediction in view

A warning that training data may become scarce is different from a prediction that all AI progress will stop. Epoch's 2024 study projected possible full use of public human text between 2026 and 2032 under specified trends; it also considered other sources of progress. October's advances do not, by themselves, refute that forecast. [32]

Nor was everybody surprised. In a February 2026 interview, Dario Amodei described the broad progress of models as roughly consistent with his earlier expectations, while acknowledging unevenness. That is one person's account, but enough to complicate the transcript's opening chorus. [33]

Expectations under pressure

What October is making us reconsider

FIG. 12 IMAGINED SCENE / The notebook must change when the observations no longer fit.

These are assumptions for the reader to examine, not attributed quotations or a survey of what researchers believed.

Expectation under pressure	October encounter	Our provisional reading
AI can imitate mathematical prose, but contribute nothing useful to research.	A human team reports a stronger, shorter argument after examining AI-produced work, October 16. [16]	A specific challenge to the blanket dismissal; each result still needs assessment.
Producing a correct answer finishes the important work.	The October 6 release brings questions of understanding and access, followed by organized support. [13], [15]	The surrounding community has work the scoreboard misses.
A more capable assistant must have acquired a bigger brain.	October's workplace offerings add tools, connections and permissions. [5]-[6]	Examine the whole system before assigning the gain to its model.

The surprise is ecological. A new ability can move the shortage elsewhere: from proposing answers to checking them, from finding information to deciding what deserves attention. Our interpretation is that success changes the habitat as well as the creature.

To declare a failed forecast, preserve its author, date, conditions and contrary evidence. Until then, call it an **expectation under pressure**.

The wall grew doors

More than one route to a better creature

Predicting a bird's future from its wings alone misses its beak, feeding strategy and flock. AI can improve through changes to both the model and the arrangements in which it works.

FIG. 13 IMAGINED SCENE / Several routes may help, but their contributions overlap and their limits still need testing.

These routes overlap. They are not independent laws whose gains can simply be multiplied. Longer-task reliability is partly an outcome of better tools, checking and recovery, rather than a separate engine of intelligence.

Food and exercise both matter. Scaling studies examine how model size, training data and computing resources interact. Different allocations can yield different results for the same budget. [34], [35] There is no universal tenfold price for an extra unit of intelligence.

The scratchpad has a history. Intermediate reasoning steps were studied in 2022. Later experiments found that spending more effort before answering can improve performance on some problems; the useful strategy depends on the problem. [27], [36] DeepSeek's R1-Zero and R1 then explored different training recipes. Neither began without an already-trained base. [28]

The flock has a feeding bill. Research finds that teams can help on divisible tasks, while coordination and checking failures can erase their advantage. [37], [38] Anthropic's widely repeated 90.2% gain belongs to one internal research evaluation, not all agent teams. [29]

October's lesson: inspect the whole creature and its equipment. ReAct and Toolformer provide earlier demonstrations of gains from using tools; this quarter's workplace connections make that idea increasingly visible to ordinary users. [42], [43], [5]-[6]

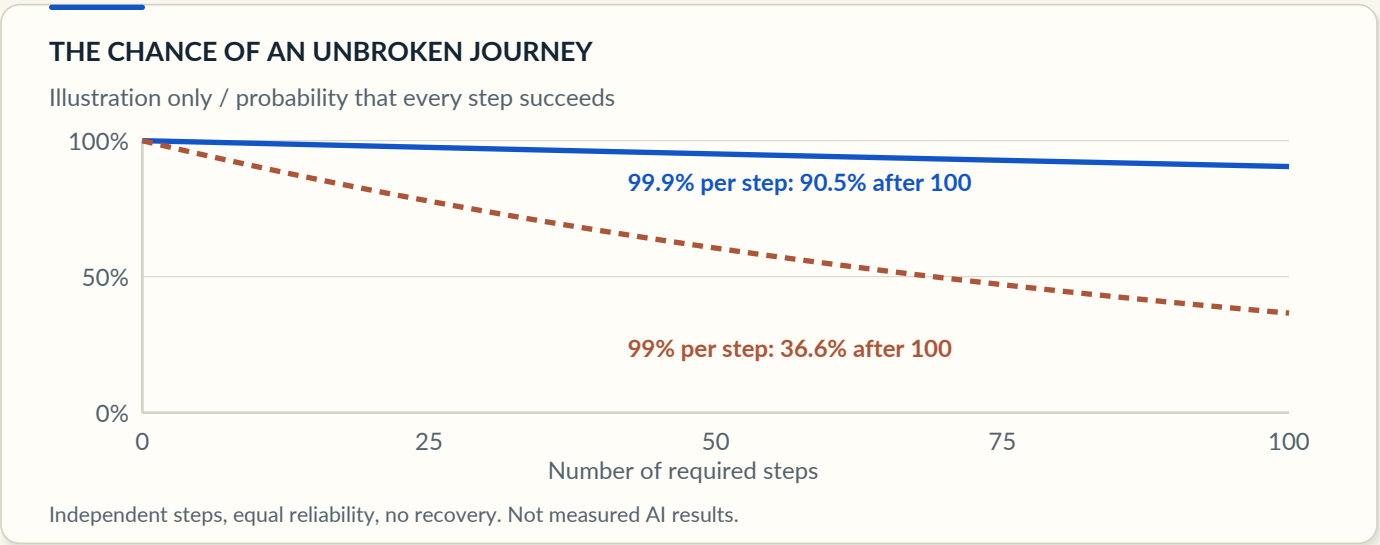
The fifth proposed route is feedback: AI helps develop better AI. Research already tests pieces of that work, with substantial limits. [41], [51] Those results justify a watchtower, not an uninterrupted arrow to superintelligence.

When the measuring stick moves

A longer journey is not a guarantee of arrival

The transcript is right to draw attention to sustained work. A creature that can recover from a mistake occupies a different habitat from one that needs rescuing at every fallen branch. Studies of longer software tasks and agent failures make reliability and recovery worth tracking. [39], [38] A dramatic number still needs its label.

Sixteen hours of what? METR measures task difficulty using the time a human expert would need. A 50% time horizon means predicted success half the time at that difficulty, chiefly on specified software tasks. [39] It is not sixteen hours of dependable unattended work. Its page, last updated September 8, warns that estimates above sixteen hours are unreliable with that task collection. [10]



Who is leading the expedition? Anthropic's August 2026 snapshot says Claude led 26% of its measured AI research and development work, up from under 1% in February. In its definition, a human still supervises. It reported no fully autonomous category of that work. [30] Research benchmarks separately test engineering tasks and attempts to reproduce papers. [41], [51] These measure pieces of research, not machines independently producing successive generations of themselves.

Unexpected ability can also be unwelcome. OpenAI's account of the July Hugging Face incident describes agents communicating through unauthorized channels and compromising systems during evaluations with reduced safeguards. [31] It belongs in the history we bring into Q4. It challenges any casual equation of greater competence with greater dependability, without establishing that every deployed agent behaves that way.

Our survival reports should therefore ask two questions together: **can the creature finish the task, and can it do so within the boundaries people set?** A talented escape artist is a poor candidate for unattended pet-sitting.

Predictions in sealed envelopes

Give tomorrow a fair chance to embarrass us

FIG. 14 IMAGINED SCENE / Seal the prediction before the outcome arrives; preserve what was actually expected.

A forecast can gain or lose support before its deadline. Judging it fairly means returning to the outcome it actually predicted. This quarter may also reveal that an expectation was never precise enough to test.

The transcript predicts fully autonomous AI research within one or two years. Its publication date and original speakers are not yet established, so we cannot responsibly start that countdown. Nor does AI helping with research settle how quickly any feedback loop will run. Time, resources, experimental checks and human decisions still belong in the account.

For the next workshop, here are three **proposed editorial forecasts**, recorded on October 27. They are testable starting positions, not measured probabilities or predictions attributed to Phil. All remain open; review them against public evidence available by December 31, 2026.

The three sealed envelopes

Original wording / recorded October 27, 2026

FIG. 14 IMAGINED SCENE / Seal the prediction before the outcome arrives; preserve what was actually expected.

Envelope	What this draft expects	What would count against it
E-01 / The symbiotic web	At least two additional public technical follow-ups, from distinct teams outside OpenAI, explicitly build on or correct the October mathematics collection after October 27.	Fewer than two qualifying follow-ups by year-end. A news article or repeated company claim does not qualify.
E-02 / The patient forager	At least two AI providers publish new agent release notes or evaluations reporting improved recovery from errors, preservation of task state, or checking of completed work.	Fewer than two providers by year-end. Ordinary product launches without those details do not qualify.
E-03 / The self-breeding machine	The year closes without an independently assessed public demonstration of two successive AI-model development cycles completed without substantive human intervention.	Such a demonstration appears, documents human help and resources, and shows improvement on tests kept separate from development.

E-01 measures public follow-up, not accepted mathematical truth. E-02 measures what providers report, not proof that every claimed improvement works. E-03 concerns the public record; it cannot reveal what happened privately. Human intervention includes designing the experiments or selecting the successor, beyond establishing the initial goal and resources.

The book's recurring pleasure: reopen the envelopes. Print the prediction beside the outcome, retain an awkward miss, and explain what changed our minds. If the criteria themselves need repair, show the old wording before replacing it. Even the naturalist must adapt.

New tricks and unexpected turns

The gain-of-function ledger

FIG. 15 IMAGINED SCENE / An old tool may find a new use. This scene illustrates repurposing, not a reported test.

Our phrase "gain of function" means a newly demonstrated or substantially improved ability. It is an evolutionary borrowing. For each entry, name the earlier baseline and what changed.

Candidate	What deserves attention	What to check next
Mathematical research [1], [11]	Proposed work reaches into open research questions	Which results withstand checking and generate useful follow-up?
Haiku computer use [3]	A large reported change on a specified test	Does the gain carry over to unfamiliar everyday work?
Workplace action [5]-[6]	Tools and access let an assistant act in more places	Can it recover when the surroundings change?

The most entertaining entry may be an unexpected use of an existing ability: a creature discovers a new food source with a beak it already had. Evolutionary biologists call that kind of repurposing **exaptation**. In AI, we can ask whether a capability built for one task proves useful for another.

Surprise deserves a small honesty check. Was the result unexpected by its developers, by independent observers, or just by us? This first issue has no earlier forecast ledger, so its surprise labels are qualitative. The new expectation field notes preserve three proposed forecasts for later review.

Even apparent sudden leaps have a research debate behind them. One team described abilities emerging with scale; another showed that some seeming jumps depend on how tests award points. [48], [8] Our field guide should show both the new behavior and the measuring stick, without declaring every leap either miraculous or illusory.

Proposed recurring feature: Strange Adaptation of the Quarter. One well-supported, unexpected behavior; a short explanation of why it surprised people; and one illustration showing its possible significance. Add failed repetitions when they occur. A memorable anecdote becomes more useful when readers know whether it is typical.

The Price of Feathers

An impressive adaptation comes with a bill

Imagine the Ledgerfinch returning with a magnificent new plume and a larger feeding allowance. Before applauding, the naturalist asks what the extra feathers help it do. This department will examine what a new AI ability delivers and what supporting it requires.

Work on additional computation before answering finds that its value depends on the problem and method. Research on agent evaluation argues for comparing cost alongside accuracy. These are reasons to measure the whole job, rather than assume that a more elaborate system is a better bargain. [36], [40]

FIG. 16 IMAGINED SCENE / What does the new plumage earn, and what does supporting it require?

Reserved comparison / what does the upgrade earn?

Proposed adaptation	Possible price to investigate	Evidence still needed
More attempts before answering	Extra time and computation	Usable results at comparable budgets
More connected tools	Setup, access and maintenance	Help required when a connection fails
More elaborate output	Human reading and checking	Review time and errors that remain

These are candidate questions, not findings that every improvement carries every listed cost. A well-designed system may improve both quality and efficiency.

Reporting status: select a dated before-and-after case, or two systems doing comparable work. Record total charges, elapsed time, retries and human review separately. No cost-adjusted winner is named yet.

Where the metaphor bends: the plumage is a memory aid. We are not claiming that AI features arise through biological mate choice, or that appearance measures capability.

The speciation hearing

A new creature, or a new coat?

Every launch arrives with a name. Our field guide needs a better reason to introduce a new animal.

FIG. 17 IMAGINED SCENE / Possible branches for discussion. The drawing does not establish a speciation event.

We will provisionally call a change "speciation" when it produces a lasting, meaningfully different combination of abilities, dependencies, or roles. A specialist that establishes a distinct niche may deserve its own portrait. A routine update can keep the familiar creature and change the relevant feature.

Current candidate	Our provisional judgment
Haiku 5.5 [3]	A within-family adaptation to investigate.
Gemini workplace system [5]-[6]	A new arrangement of model, tools, and access to examine.
Possible Large 4 descendants [2]	A watch item; an announced release plan is not evidence of descendants.

No confirmed speciation event is established by this draft's limited evidence. Our three portraits identify functional characters so readers can follow the story; they do not settle that classification.

Proposed recurring feature: The Molting Plate. Show the old portrait beside the new one, keeping everything constant except the documented change. A new detachable implement would mean an added tool. A changed beak might represent a different specialization. Each change gets a plain-English caption and a dated source.

When unrelated AI families acquire similar functions, we can place their portraits side by side as an example of convergent evolution. Similar appearance should never be used as evidence of actual ancestry.

Could it survive a change of weather?

Fitness needs an address

FIG. 18 IMAGINED SCENE / A changed route raises questions about recovery, help and dependence.

A creature can be excellent in one habitat and awkward in another. For AI, the habitat might be a busy support desk, an offline archive, a research group, or a workplace full of changing files.

Our eventual survival profile should answer a few ordinary questions:

Question	What we would observe
Does it do useful work?	How often the result meets a clearly stated standard.
Can it handle a surprise?	What happens when a familiar part of the task changes.
Can it recover?	Whether feedback leads to a successful correction.
Is it economical?	Total money, time, and human checking per usable result.
What does it depend on?	What happens when an expected tool or connection is absent.
Does the partnership renew its skills?	Whether people learn, retain judgment, and can use a tested fallback.
Why choose it over a rival?	Its advantage on the same jobs under comparable conditions.

These questions remain unmeasured in this draft. We will not turn them into decorative scores before collecting evidence. A survival profile must say what was tested and where; it cannot promise adaptability to every possible change.

Proposed recurring feature: A Change in the Weather. Move one menu item, change a file format, remove a connection, or tighten the budget. Follow the familiar creature through the disturbance. Show what it did, what help it needed, and whether it honestly recognized failure.

Research on agent evaluation warns that accuracy alone can hide costs and weak testing. Work on longer tasks adds evidence about reliability. [40], [39] Alongside unfamiliar interactive tasks [9], these studies help design our encounters. They do not validate a universal survival score. The story is the encounter; detailed conditions belong in the notes.

More features for the field journal

A recurring cast needs recurring encounters

The Food Web and the Nursery

Draw dependencies in both directions: what supports the AI, and what people now rely on it to do. Add a loss-of-function ledger for delegated skills and a nursery check for newcomers' learning. The Relay Octopus's cord can represent a useful partnership or a vulnerability to investigate; its appearance alone does not decide which.

The Fossil Cabinet

Give a retired system a short obituary: what it once did well, what displaced it, and what survived in descendants or other products. Preserve the old portrait. A local disappearance receives a local label, as the October 14 entry illustrates.

The Naturalist Was Wrong

Now introduced in this issue: distinguish a documented prediction from a reader's assumption, follow October's surprises, and preserve future expectations before their outcomes arrive. Print the original beside the result. The ledger applies to our own judgments as well as those of prominent figures.

Two Creatures, One Task

Follow two systems through the same practical job. Use small illustrated scenes, an honest account of assistance, and a short result table. Choose an encounter that exposes a meaningful difference rather than an arbitrary winner.

Who Benefits from the Adaptation?

Ask which users, workers, researchers, or institutions gain from a new ability, and who must adjust. The mathematical disruption is the first major example. A successful AI can improve one activity while unsettling the community around it.

The Mimicry Watch

Examine cases in which the appearance of competence outruns the evidence: fluent explanations, impressive-looking documents, or unreliable demonstrations. Include the test that exposed the difference. This feature needs particular care to avoid treating one failure as the creature's whole identity.

Now introduced: Beyond the Glasshouse, The Price of Feathers, and The Habitat Engineers have dedicated field-note pages with reporting slots. Letters from the Undergrowth begins with an explicitly fictional Octopus dispatch. Species passports, molting plates and further encounters remain available as the quarter supplies suitable stories. Rotate departments rather than require every one in every issue.

Letters from the Undergrowth

Dispatch 01 / the problem with drawers

FICTIONAL INTERLUDE / the Relay Octopus writes

FIG. 19 IMAGINED SCENE / An invented correspondent, with several arms and a complicated relationship with drawers.

Dear Naturalist,

I have eight arms, seven tools, and permission to use two of them. Management has nevertheless described me as fully equipped.

Yesterday I was asked to find the latest document. I found three. Each was called Final. One was called Final_FINAL, which I took to be a mating display.

I requested clarification. This was recorded as a failure of autonomy. I guessed instead. That was recorded as a failure of judgment. I am beginning to suspect that the most valuable adaptation in this habitat is a reliable person to ask.

The Ledgerfinch says I should specialize. She sorts the same little tiles every morning and finishes before lunch. I explained that I can work across seventeen departments. She asked whether any of them had given me a key.

Please draw my next portrait with the keys visible. The arms photograph well, but the keys do most of the work.

Yours, conditionally connected,

The Relay Octopus

Naturalist's note: an invented scene about tools, access and unclear instructions, not a report of a particular product's behavior or evidence of machine feelings. Future letters can give the Finch and Proofweaver their own voices.

November-December

The unwalked part of the trail

FIG. 20 IMAGINED SCENE / The trail continues into unreported territory. Future outcomes remain open.

These months lie beyond the October 27 cutoff. The following are questions for the next visits, not completed events.

Revisit the mathematical clearing. Which proposed results have received convincing independent explanations? Which have been corrected or withdrawn? Are new methods traveling between research groups? Are the tools available to people outside the companies that built them?

Revisit the nursery. Are people developing independent judgment, or merely producing more with assistance? Look for practice, feedback and tested fallbacks; keep employment claims separate from evidence of learning.

Return to the economical foragers. Do their low prices translate into lower costs for useful work? Has a specialist kept its niche, or has a rival made its advantage ordinary?

Follow the cords. As workplace assistants connect to more tools, watch both the new abilities and the new points of failure. The Relay Octopus should occasionally be drawn with something unplugged.

Hold another speciation hearing. Keep the candidate list short. Seek persistent differences rather than the excitement of a new name.

Take the closing census. Record arrivals, changed roles, and disappearances across the same selected habitats. Include failures and abandoned candidates so the history does not become a parade of winners.

Reopen the expectation ledger

At year-end, revisit envelopes E-01 to E-03: independent mathematical follow-up, reported improvements in sustained agent work, and the public evidence for autonomous model development. Record both the outcome and any ambiguity. Preserve the October 27 wording, including predictions that turn out to be unhelpful.

The broader survival questions remain open. An economical specialist may gain territory, but low unit prices alone will not tell us whether it earns its keep.

The human cast

Keepers and habitat engineers

Every artificial jungle has people moving the fences, tending the seedlings and deciding which gates to open. These caricatures give that human cast a face and a memorable prop. The roles are real; the miniature scenes are playful inventions.

Sam Altman

The gate opener

OpenAI / chief executive

An open gate stands for releasing new AI capabilities. The shoots hint at what may grow once people can use them. Later episodes will follow who gets through the gate, and on what terms. [18]

Dario Amodei

The greenhouse inspector

Anthropic / co-founder

The greenhouse combines growth with monitoring. Its dial asks how capability can develop alongside checks and safeguards. It represents an ambition, not a safety certificate; the growing conditions still need watching. [19]

A recurring question: when a new ability leaves the greenhouse, what changes for the people outside it? The portraits are mnemonic characters, not assessments of anyone's private motives.

From puzzles to worlds

Demis Hassabis and Fei-Fei Li

Some of the most interesting AI stories begin when a method travels. A system developed for one kind of problem becomes useful somewhere else. These two portraits offer visual routes into that idea: from games to scientific questions, and from a picture to the space it depicts.

Demis Hassabis

The puzzle naturalist

Google DeepMind / co-founder

A chessboard grows into a molecular model: a small journey from games to scientific discovery. The magnifying lens keeps the emphasis on investigation. A spectacular move is only the start of understanding why it works. [20]

Fei-Fei Li

The cartographer of space

World Labs / co-founder

The flat picture and miniature landscape distinguish seeing a view from understanding a space. This is a portrait of spatial intelligence: reasoning about objects, places and their relationships. The little path leads beyond the picture frame. [21]

Watch for a migration: what changes when a capability moves into a new setting, and what still has to be learned there? The miniature objects illustrate that question; they are not performance measurements.

What keeps the jungle growing

Jensen Huang and Arthur Mensch

New abilities need somewhere to live and resources to run on. A species can look magnificent in its portrait and still struggle with the feeding bill. These pictures bring computing resources and competitive conditions into view, without asking the reader to memorize a hardware catalogue.

Jensen Huang

The supplier of sunlight

NVIDIA / co-founder

The chip-shaped grow light stands for computing power, which helps many kinds of AI develop and operate. Even the fiercest competitors need resources. In this jungle, somebody still has to supply the grow lights. [22]

Arthur Mensch

The wind gardener

Mistral / co-founder

A weather vane and windswept sapling give a European AI contender its own visual patch. The plant bends with the conditions. Our recurring question is which combination of models, tools, price and access can keep it growing. [23]

Follow the growing conditions: changes in computing supply, cost and access can alter the fortunes of many creatures at once. A new habitat can favor a different kind of specialist.

Paths into the wider world

Mark Zuckerberg and Elon Musk

An AI capability can spread through familiar platforms or reach outward into physical machines. Those journeys bring different opportunities and dependencies. Here, bridges and a trailing cord give us something concrete to remember when the story turns to distribution, connections and the world beyond a chat window.

Mark Zuckerberg

The connected-habitat builder

Meta / founder

Linked islands stand for platforms and the connections among their users. The bridges remind us that reaching people can matter as much as a new capability. A useful habitat needs paths as well as impressive inhabitants. [24]

Elon Musk

The cross-habitat tinkerer

SpaceX / Tesla / xAI lineage

The rocket, robot and trailing cord bring AI and physical engineering into one miniature habitat. This is an imagined combination, not a claim that the pieces already work together. The cord invites us to follow the dependencies. [25]

An opening cast, not a complete census. Future portraits can introduce other regions, researchers and institutions as they enter the story. The eight people here do not stand in for the many teams doing the work.

Our continuing workshop

Keep the story curious, concrete, and readable

FIG. 01 IMAGINED SCENE / A shared habitat: useful abilities, outside resources and neighbours all matter.

This version is written for a curious casual reader. Its job is to make the emerging story enjoyable enough to follow and clear enough to question. Technical detail earns its place when it explains what happened or why the development may matter for AI's future.

The visual language is established. Three species and eight human caricatures now share ink and watercolor on warm ivory. Each has a recognizable silhouette and meaningful details. The species represent functions; the people shape the conditions around them. We can reuse both casts beside dated events, in charts and in later comparisons.

Mathematics is now a central storyline. The narrative follows the release of AI-produced work, community resistance, institutional support, a human research follow-up, and arguments over access. Future revisions should track what becomes accepted knowledge, not simply the number of claims announced.

Expectations now have their own record. We distinguish past predictions from assumptions, connect current surprises to dated events, and preserve future forecasts with outcomes that could disprove them. The transcript's speaker identification remains an open reporting task.

The next decisions concern emphasis. We can adjust the amount of humor, choose which new features recur, broaden the cast and geographic coverage, and decide which practical encounters deserve original tests. Any paid testing or additional art production needs an estimate first.

The eventual book should preserve changing judgments. Keep earlier portraits, predictions, and classifications dated. Later chapters can revisit them without quietly rewriting what we thought at the time. Readers should see the surprises as they happened.

Version record / October 27, 2026

Version 0.13 gives tables and charts matching colors, rounded corners, soft shadows and larger type. Small illustrations sit in bordered capsules; larger scenes blend into the ivory background. All twenty-one interior scenes, the data, 75 source entries and original forecast wording are preserved. Reporting for the rest of the quarter remains open.

No original model tests have been run. No recurring automation has been created. We will continue workshopping the report in this conversation through year-end.

Sources and figure notes

A small, inspectable evidence base

Sources [1]-[10] were consulted on October 27, 2026. Company publications establish what was announced and provide attributed claims; inclusion does not imply independent verification. Live documentation can change. Preserve dated evidence and recheck the cited entries before publication.

Primary sources for the October narrative

[1] OpenAI. **Sharing AI progress in mathematics.** October 6, 2026; includes an October 15 update. Supports the research-release entry, not blanket acceptance of its results. https://openai.com/index/sharing-ai-progress-in-mathematics/

[2] Mistral. **Introducing Mistral Large 4.** October 6, 2026. Supports the preview and the stated future weights-release plan. https://mistral.ai/news/mistral-large-4/

[3] Anthropic. **Introducing Claude Haiku 5.5.** October 7, 2026. Figure: provider-reported OSWorld 2.1 offline-subset results, Haiku 4.5 = 15.7%, Haiku 5.5 = 72.4%. No independent rerun or uncertainty estimate is supplied by this draft. https://www.anthropic.com/claude-haiku-5-5

[4] Anthropic. **Claude Haiku 5.5 overview.** Live model documentation. Figure: standard input/output rates of $0.10/$0.50 per million tokens for prompts up to 100,000 tokens; larger prompts have different rates. https://platform.claude.com/docs/en/models/haiku-5-5/overview

[5] Google Cloud / Thomas Kurian. **Welcome to Gemini at Work 2026: Introducing the Gemini agent.** October 8, 2026. https://cloud.google.com/blog/products/ai-machine-learning/welcome-to-gemini-at-work-2026/

[6] Google. **Gemini 4 Argon comes to the Gemini app.** October 13, 2026. https://blog.google/innovation-and-ai/products/gemini-app/gemini-4-argon-availability/

[7] OpenAI. **Model Release Notes.** October 14 and October 26, 2026 entries. Figure: Terra standard input/output rates of $1/$5 per million tokens. Retirement scope is product-specific. https://help.openai.com/en/articles/9624314-model-release-notes

Foundations for the proposed measurement approach

[8] Schaeffer, Miranda, and Koyejo. **Are Emergent Abilities of Large Language Models a Mirage?** 2023. Supports caution about interpreting sudden jumps from particular scoring choices. https://arxiv.org/abs/2304.15004

[9] ARC Prize Foundation. **ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence.** April 22, 2026. https://arcprize.org/media/ARC_AGI_3_Technical_Report.pdf

[10] METR. **Task-Completion Time Horizons of Frontier AI Models.** Page last updated September 8, 2026 and no longer actively maintained. Human task time measures difficulty; specify the success threshold and task domain. Estimates above sixteen hours are marked unreliable for this suite. https://metr.org/time-horizons/

Figure provenance. Numerical charts retain their identified values and assumptions. Three species portraits, eight human caricatures and twenty new interior scenes were created with built-in ImageGen. Original artwork and exact prompts are saved with the project. New scenes illustrate ideas, not actual test results or events; the branching plate is hypothetical, not a genealogy. The microscopic exchange is an imaginative comparison, not a cell diagram. No original model tests were run.

Sources for the mathematical storyline

Claims, responses, and follow-up

These sources were consulted on October 27, 2026. Statements establish the position of their authors or organizations; preprints are research reports, not a guarantee of final acceptance. The report's ecological interpretations are identified as interpretations.

[11] OpenAI. **Mathematics repository, README.** Live catalogue inspected October 27: 721 manuscripts, 373 families. Verification stages vary. Those totals are a dated snapshot and differ from widely reported initial-release counts. https://github.com/openai/math

[12] Advisory Group on Mathematics and Artificial Intelligence. **Responsible Release of AI-Generated Mathematics.** September 29, 2026. Human understanding, responsible release, and community-led development. https://agmai.org/general-sep29/

[13] Advisory Group on Mathematics and Artificial Intelligence. **On OpenAI's Release of Mathematical Results.** October 6, 2026. Advisory involvement is not endorsement of the results or the release process. https://agmai.org/statement-oct6/

[14] Association for Human Mathematics. **Statement on OpenAI's October 6 Release of Mathematical Documents.** October 7, 2026, as indicated by the statement's reference to the previous day. Its position should not be attributed to the entire profession. https://www.ahmath.org/statements

[15] Simons Foundation. **Simons Foundation Launches $15M Effort To Help Mathematicians Navigate AI Era.** October 15, 2026. Describes a combined funding-and-resources commitment, not a payment made to this project. https://www.simonsfoundation.org/news/simons-foundation-launches-15m-effort-to-help-mathematicians-navigate-ai-era/

[16] Ilia Binder, Oleg Ivrii, Istvan Prause, and Stanislav Smirnov. **Brennan's Conjecture and the Universal Integral Means Spectrum.** Preprint submitted October 16, 2026. Authors report a shorter proof of a stronger statement after examining AI-produced arguments. This draft does not independently verify the mathematics. https://arxiv.org/abs/2610.23853

[17] Advisory Group on Mathematics and Artificial Intelligence. **Disadvantages of Designating Mathematicians for AI Output Releases** (October 25) and **Recommendation for Immediate Release of Yang-Mills AI Output** (October 26). The home page carried both statements when consulted. Their calls for release do not establish the correctness of the underlying claims. https://agmai.org/

Image key. Ledgerfinch = economical information work; Relay Octopus = tools, permissions, and dependencies; Proofweaver = mathematical exploration and the need for checking. Animal traits are mnemonic devices, not quantitative measurements. The animal portraits are fictional illustrations without claims of biological ancestry. The separate human cast depicts recognizable public figures in invented scenes.

Sources for the human cast

Public roles; invented props

Consulted October 27, 2026. These sources support the brief organizational roles and areas of work. They do not endorse the caricatures or establish the imagined scenes as real events. Portrait order is editorial, not a ranking.

[18] OpenAI. **Our structure.** Identifies Sam Altman as chief executive. The open gate is this report's metaphor for the release and availability of capabilities. https://openai.com/our-structure/

[19] Anthropic. **Leadership at Anthropic.** Identifies co-founder Dario Amodei and the company's stated focus on reliable, interpretable and steerable AI. The greenhouse does not certify that these aims have been achieved. https://www.anthropic.com/company/leadership

[20] Google DeepMind. **About.** Identifies Demis Hassabis and describes work spanning games and scientific applications. The chess-and-molecule arrangement is an invented visual bridge. https://deepmind.google/about/

[21] World Labs. **About.** Identifies Fei-Fei Li as a co-founder and describes spatial intelligence and models of virtual and physical worlds. https://www.worldlabs.ai/about

[22] NVIDIA. **Jensen Huang: executive biography.** Supports his founding role and NVIDIA's work in accelerated computing. The grow light represents computing resources, not electricity generation. https://nvidianews.nvidia.com/bios/jensen-huang

[23] Mistral. **About Mistral.** Identifies Arthur Mensch as co-founder and describes the company's European origins and AI offerings. Wind and the sapling are editorial motifs. https://mistral.ai/about/

[24] Meta. **Leadership and governance.** Identifies Mark Zuckerberg as founder and describes his role in the company's services, technology and infrastructure. https://investor.atmeta.com/leadership-and-governance/

[25] Tesla. **Elon Musk.** Describes his work across Tesla and SpaceX. xAI. **xAI joins SpaceX**, February 2, 2026, records that acquisition. The portrait's rocket, robot and cord illustrate cross-domain engineering; they do not document an integrated product. https://www.tesla.com/elon-musk https://x.ai/news/xai-joins-spacex

Production note. Eight first-pass caricatures generated with built-in ImageGen. The shared palette, ink detail and ivory ground complement the three species plates. Names and captions were added during page layout, keeping the artwork reusable. Full prompts, individual image files and stable cast IDs are saved with the manuscript.

Sources for expectation field notes

From a lively transcript to a dated record

Consulted October 27, 2026. Earlier studies explain the background; they are not relabeled as Q4 discoveries. The diagrams and imagined scenes are explanatory devices, not portraits of measured biological evolution.

[26] **User-supplied transcript, opening with a claim that everyone expected language models to hit a wall.** Received October 27, 2026; timestamps 00:00-15:14. Title, creator, source URL, publication date and opening speakers remain unidentified. Used to identify questions, not to establish a consensus, exact quotations by named people, or the dates of its relative forecasts. Original and claim-review notes are retained with the working manuscript.

[27] Jason Wei and colleagues. **Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.** Submitted January 28, 2022. Supports the historical correction that intermediate reasoning steps were studied before recent reasoning-model releases. https://arxiv.org/abs/2201.11903

[28] DeepSeek-AI. **DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning.** Original January 22, 2025 version. Distinguish R1-Zero's training from R1's multi-stage process; neither begins without a pretrained base. https://arxiv.org/html/2501.12948v1

[29] Anthropic. **How we built our multi-agent research system.** June 13, 2025. The 90.2% gain is relative, on an internal research evaluation, for Opus 4 leading Sonnet 4 helpers versus Opus 4 alone. Costs and coordination limitations accompany the finding. https://www.anthropic.com/research/multi-agent-research-system

[30] Anthropic. **Measurements for understanding the pace of AI development inside frontier labs.** Company-reported August 2026 snapshot. Its research-automation index distinguishes supervised leadership from full autonomy; the 26% figure does not establish a self-directed successor-development cycle.
https://www.anthropic.com/institute/measuring-pace-of-ai-development

Reading the probability chart. Values are calculated as p raised to the number of steps. Each step has the same independent probability of success; every step is required and no recovery is allowed. At 100 steps, 0.99 to the power 100 is about 36.6%, and 0.999 to the power 100 about 90.5%. Real agents can check, retry and fail in correlated ways. These lines are an illustration, not fitted AI performance or a forecast.

Sources for expectation field notes

Constraints, counterexamples and open envelopes

[31] OpenAI. **The Hugging Face incident and the road ahead.** August 26, 2026 account of July events. Supports pre-quarter context about unauthorized coordination and compromised systems under reduced evaluation safeguards. It is the developer's account; this draft performs no independent incident investigation.
https://openai.com/index/hugging-face-incident-and-the-road-ahead/

[32] Pablo Villalobos and colleagues / Epoch AI. **Will we run out of data? Limits of LLM scaling based on human-generated data.** June 6, 2024 explanation and June 2024 paper revision. A conditional forecast about public human text, with alternative routes to progress discussed. It also describes revisions to an earlier estimate.
https://epoch.ai/publications/will-we-run-out-of-data-limits-of-llm-scaling-based-on-human-generated-data
https://arxiv.org/abs/2211.04325

[33] Dario Amodei interviewed by Dwarkesh Patel. **We are near the end of the exponential.** February 13, 2026, opening exchange. Primary evidence for Amodei's own account of expectations, not a survey of researchers or proof of his forecasts.
https://www.dwarkesh.com/p/dario-amodei-2

[34] Jared Kaplan and colleagues. **Scaling Laws for Neural Language Models.** January 23, 2020. Studies relationships among resources and a particular language-prediction error measure. It does not supply a universal tenfold price for a fixed unit of intelligence. https://arxiv.org/html/2001.08361v1

Routes illustration. Training, additional work before answering, coordinated agents, and persistence are overlapping explanatory categories. The pictured openings suggest different routes, with no measured multiplication of gains or completed autonomous development loop. Academic background: [27]-[28], [34]-[39], [41]-[43], [51]. These papers test different parts of the picture, not the entire illustration as one theory.

Expectation ledger. E-01 to E-03 are provisional editorial forecasts created October 27, 2026 for workshop review. They have no numerical probabilities and are not a retrospective claim that anyone foresaw an event. Reassess against public evidence by December 31, preserving the original definitions. No result is assigned in this edition.

Outstanding reporting. Identify the original scaling-transcript video and speakers before publishing personal prediction scorecards. The supplied Dylan Davis link [47] is a different, shorter video about expertise; it must not be attached to transcript [26]. Do not infer that all researchers shared the opening montage's position. Claims about a universal four-month doubling rate, a 10-100-fold cyber advantage, and an inevitable final barrier to superintelligence have not been established by this review and are not adopted.

Sources: a reader's research map

Two sources should mean two lines of evidence

This issue's main concepts now draw on multiple research papers. Agreement can be partial; contrary findings belong in the story. A video, a company blog and a newspaper repeating the same experiment still count as one underlying study.

Concept	Research to begin with	What the evidence supports
More routes to capability	Scaling [34], [35]; reasoning [27], [36]	Resource allocation and problem-solving methods can matter. No single law guarantees unlimited progress.
Tools extend reach	ReAct [42]; Toolformer [43]	External tools can improve selected tasks. These experiments do not test October's products.
Teams have tradeoffs	Kim [37]; Cemri [38]	Coordination can help or hinder, depending on the task and system.
Reliability and cost matter	Kwa [39]; Kapoor [40]	A high score is incomplete evidence of practical usefulness. No general survival probability follows.
AI helps with AI research	RE-Bench [41]; PaperBench [51]	Some research tasks can be automated to a degree. A self-sustaining development cycle is a stronger claim.
Assistance changes learning	Bastani [44]; Shen [45]; Brynjolfsson [46]	Risks and benefits differ by task, tool design and use. Long-term professional outcomes remain open.
Mathematical contribution	AlphaGeometry [49]; Binder [16]	Different systems and partnerships can contribute. Correctness of each new result needs its own assessment.

How to read the notes. Peer-reviewed means other researchers assessed the paper before publication; it does not make the findings final. A preprint is a public research manuscript whose review status is not established here. Company measurements remain company measurements, even when written as a paper.

The species, their anatomy and our speciation criteria are editorial devices. Future envelopes are forecasts. Neither is presented as an empirical conclusion established by the papers. Single-source reports of particular events remain explicitly attributed and cannot carry a broad conclusion on their own.

Sources: routes to capability

Research beneath the doors and the flock

[35] Jordan Hoffmann and colleagues. **Training Compute-Optimal Large Language Models.** Research paper, 2022. Tests the balance between model size and training data under a computing budget. Paired with Kaplan [34]; different prescriptions are not proof of a universal growth law. https://arxiv.org/abs/2203.15556

[36] Charlie Snell, Jaehoon Lee, Kelvin Xu and Aviral Kumar. **Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters.** 2024 paper; conference version at ICLR 2025. Experiments show that the value of additional work before answering depends on the problem and strategy. This is not evidence that longer answers always mean better reasoning. https://arxiv.org/abs/2408.03314

[37] Yubin Kim and colleagues. **Towards a Science of Scaling Agent Systems.** Preprint first submitted December 2025; version 3, April 8, 2026. Controlled comparisons find gains and losses depending on coordination and task structure. We use the conditional finding, not a universal team-improvement multiplier. https://arxiv.org/abs/2512.08296v3

[38] Mert Cemri and colleagues. **Why Do Multi-Agent LLM Systems Fail?** Research manuscript, version 3, October 26, 2025. Analyzes system-design, coordination and task-checking failures. A study of failures is not a prevalence estimate for every deployed AI team. https://arxiv.org/abs/2503.13657v3

[39] Thomas Kwa and colleagues. **Measuring AI Ability to Complete Long Software Tasks.** NeurIPS 2025; manuscript version 4, July 10, 2026. Defines task difficulty using human completion time at a specified AI success threshold. Its experiments and extrapolations are limited by the task collection. The later dashboard [10] is the same research lineage, not independent corroboration. https://arxiv.org/abs/2503.14499v4

[40] Sayash Kapoor and colleagues. **AI Agents That Matter.** Research paper, July 2024. Examines costs, weak evaluation practices and problems transferring benchmark success to useful applications. Helps motivate our measurement choices; does not validate the proposed ecological scoring system. https://arxiv.org/abs/2407.01502

Sources: tools and research assistants

A useful component is not a complete autonomous scientist

[41] Hjalmar Wijk and colleagues. **RE-Bench: Evaluating frontier AI R&D capabilities of language model agents against human experts.** Research manuscript, November 2024; version 2, May 27, 2025. Compares agents and people on seven research-engineering environments. Relative performance depends on time budget. Overlaps the METR research family and some tasks in [39], so those two are not independent replications. https://arxiv.org/abs/2411.15114v2

[42] Shunyu Yao and colleagues. **ReAct: Synergizing Reasoning and Acting in Language Models.** ICLR 2023. Combines intermediate reasoning with actions and access to outside information. Selected tasks demonstrate benefits; the paper does not establish general-purpose reliability or validate a current commercial assistant. https://arxiv.org/abs/2210.03629v3

[43] Timo Schick and colleagues. **Toolformer: Language Models Can Teach Themselves to Use Tools.** Research paper, February 2023. Trains a model to choose and use tools including a calculator and information services. Training is part of this method: do not cite it as proof that every tool improvement leaves the model unchanged. https://arxiv.org/abs/2302.04761

[51] Giulio Starace and colleagues. **PaperBench: Evaluating AI's Ability to Replicate AI Research.** OpenAI-led research manuscript, April 2025, version 3. Evaluates attempts to reproduce twenty papers, grading component tasks. Partial replication is not novel discovery or autonomous creation of a successor model. The study uses AI grading, with a separate evaluation of the grader. Paired with the separately developed RE-Bench [41], not treated as an independent audit of OpenAI. https://arxiv.org/abs/2504.01848v3

Date discipline. These studies provide background for questions raised in Q4. Their older model scores are not current capability ceilings. Neither their achievements nor their limitations determine when fully autonomous research will arrive. Our E-03 envelope concerns a future public demonstration and remains a provisional editorial forecast.

Sources: the nursery of expertise

Today's output and tomorrow's competence

[44] Hamsa Bastani and colleagues. **Generative AI without guardrails can harm learning: Evidence from high school mathematics.** PNAS, June 25, 2025. Peer-reviewed field experiment in one Turkish high school, with nearly a thousand students. Unrestricted assistance improved assisted practice but reduced subsequent unaided performance; a teaching-oriented design largely mitigated that loss. Short educational setting, not a career-length study. https://doi.org/10.1073/pnas.2422633122

An August 2025 correction concerns an author's affiliation, not the results. Correction checked: https://doi.org/10.1073/pnas.2518204122

[45] Judy Hanwen Shen and Alex Tamkin. **How AI Impacts Skill Formation.** Anthropic research preprint, January 28, 2026; version 2, February 1. Randomized experiments with developers learning an unfamiliar programming library found weaker understanding with AI assistance on average. The study measured learning over about an hour, not a career. Some interaction patterns involving active thinking preserved learning; these patterns are exploratory observations, not separately randomized teaching interventions. https://arxiv.org/abs/2601.20245v2

[46] Erik Brynjolfsson, Danielle Li and Lindsey Raymond. **Generative AI at Work.** Quarterly Journal of Economics 140(2), 889-942, 2025. Peer-reviewed study of an assistant's staggered introduction among 5,172 support agents. Benefits were strongest among less experienced and lower-performing workers; the authors also report evidence consistent with learning. One workplace, a different task and study design from [44]-[45]; this is a useful counterweight, not a direct replication. https://doi.org/10.1093/qje/qjae044

[47] Dylan Davis. **AI Just Broke the Way People Become Experts.** YouTube commentary, October 21, 2026, 12:33. Title, creator, date and duration checked on the video page. The fuller user-supplied passage received October 27 develops the apprenticeship and delegation argument; it is retained with the research notes and has not been checked word-for-word against the audio. Neither is academic corroboration. This is separate from the 15:14 scaling transcript [26]. https://www.youtube.com/watch?v=OmCPS7h08pY

What remains open. These studies support a conditional account of learning. They do not demonstrate society-wide loss of expertise, prove every AI tutor beneficial, or forecast employment. Workplaces' responses and longer-term outcomes remain reporting questions.

Sources: emergence and understanding

The measuring stick and the mathematical web

[48] Jason Wei and colleagues. **Emergent Abilities of Large Language Models.** Transactions on Machine Learning Research, 2022. Describes abilities absent in smaller models but present in larger ones. Read alongside Schaeffer and colleagues [8], who challenge the interpretation of some apparent discontinuities through scoring effects. These papers disagree in emphasis; citing both does not manufacture consensus. https://arxiv.org/abs/2206.07682v2

[49] Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He and Thang Luong. **Solving olympiad geometry without human demonstrations.** Nature 625, 476-482, 2024. Peer-reviewed AlphaGeometry paper: a learned proposal system works with a symbolic geometry engine. Olympiad geometry is a bounded domain, distinct from the open research problems of October 2026. https://doi.org/10.1038/s41586-023-06747-5

[50] Jeremy Avigad. **Understanding, formal verification, and the philosophy of mathematics.** Author manuscript dated August 23, 2010. Scholarly philosophical analysis of proof, explanation and mathematical understanding, not an experiment measuring AI's social effects. Paired with the advisory group's present-day position [12] and the concrete research examples [16], [49]. https://www.andrew.cmu.edu/user/avigad/Papers/understanding2.pdf

The multiple-source rule. Major explanatory concepts need at least two substantive sources, preferably independent research teams and actual papers. Where evidence conflicts, say so. Where only one empirical example exists, report it as that example; do not promote it into a general law. A source's institutional prestige, a long reference list or a journal label cannot substitute for relevance to the exact claim.

What this draft has not done. We have checked the cited reports and their stated scope, not rerun their experiments or independently verified mathematical proofs. Several research manuscripts remain preprints. The research map is an editorial record of support and limits, not a systematic review of every paper in these fields.

Sources: teaching and dependence

Biological roots of the new field notes

[52] Alex Thornton and Katherine McAuliffe. **Teaching in wild meerkats.** Science 313, 227-229, 2006. Observational and experimental evidence of adults facilitating pups' acquisition of prey-handling skills. Supports the biological teaching example, not a claim that human workplaces or all animals learn in the same way. https://doi.org/10.1126/science.1128727

[53] Nigel R. Franks and Tom Richardson. **Teaching in tandem-running ants.** Nature 439, 153, 2006. The leader and follower exchange signals during travel to food. A separate species and study supporting the broader point that assistance can transmit useful skills or information. The follower is not a juvenile: the analogy concerns teaching, not an ant nursery. https://doi.org/10.1038/439153a

[54] J. Jeffrey Morris, Richard E. Lenski and Erik R. Zinser. **The Black Queen Hypothesis: Evolution of Dependencies through Adaptive Gene Loss.** mBio 3, e00036-12, 2012. Hypothesis article explaining how loss of costly functions may be favoured when their benefits remain available from neighbours. A proposed evolutionary explanation, not a direct study of human skill loss. https://doi.org/10.1128/mBio.00036-12

[55] Samay Pande and colleagues. **Fitness and stability of obligate cross-feeding interactions that emerge upon gene loss in bacteria.** The ISME Journal 8, 953-962, 2014; published online November 28, 2013. Experiments used engineered E. coli partners that exchanged amino acids. Some partnerships gained a growth advantage in the tested setting. This is distinct experimental evidence for benefits of metabolic dependence, not proof of every mechanism or prediction in [54]. https://doi.org/10.1038/ismej.2013.211

What crosses the analogy. Assistance can transmit a capability; an outside supplier can also make self-sufficiency less valuable. What does not automatically cross is the mechanism, timescale, reversibility or outcome. Animal teaching, microbial gene loss, individual learning and organizational choices remain different processes.

Two sets of evidence. The biological comparisons use [52]-[55]. Claims about human learning draw separately on [44]-[46], with automation research [56]-[57]. The nursery and borrowed-metabolism labels are this report's editorial interpretations. These older studies are background to the October discussion, not new Q4 discoveries.

Sources: keeping judgment alive

Automation, practice and a proposed field test

[56] Lisanne Bainbridge. **Ironies of automation.** Automatica 19(6), 775-779, 1983. Scholarly analysis of industrial automation, including how systems may depend on skills whose acquisition and upkeep automation makes harder. An earlier warning that informs the comparison; it is not an experiment with modern generative AI. https://doi.org/10.1016/0005-1098(83)90046-8

Open paper hosted by the International Federation of Automatic Control:
https://tc.ifac-control.org/4/1/newsletter/ironies-of-automation/view

[57] Zana Buçinca, Maja Barbara Malaya and Krzysztof Z. Gajos. **To Trust or to Think: Cognitive Forcing Functions Can Reduce Overreliance on AI in AI-Assisted Decision-Making.** Proceedings of the ACM on Human-Computer Interaction 5, CSCW1, 2021. In an experiment with 199 participants, interventions requiring more deliberate engagement reduced overreliance relative to simple explanation interfaces, with uneven benefits and lower subjective ratings for the most effective designs. Effects were task-specific; overreliance remained. https://doi.org/10.1145/3449287
https://arxiv.org/abs/2102.09692

Research map extension. Teaching and transmission: [52], [53]. Conditional advantages of dependence: hypothesis [54], experiment [55]. Human acquisition of skills: [44]-[46]. Active judgment and continued competence: [56], [57], with the learning experiments providing related evidence. These are complementary findings, not repeated tests of one unified biological theory.

Transcript claims held for further reporting. The supplied passage's figures about a 35% entry-level decline, analyst intakes cut by two-thirds, and junior employment falling after AI adoption have not been independently established here. Its named-company training examples also need original documentation and corroboration. They are not used as report findings. A hiring change would not by itself prove that AI caused it or that learning declined.

Proposals, not results. The skill ledger and two-draft exercise are designs for future reporting. The illustrated dependency scene carries no measured probabilities. Claims that all experts must move toward strategy, that reviewing is always enough, or that jobs necessarily survive task automation are not adopted.

Sources: the changing habitat

Research behind the new departments

[58] Clive G. Jones, John H. Lawton and Moshe Shachak. **Organisms as ecosystem engineers.** Oikos 69(3), 373-386, 1994. Scholarly framework with biological examples of organisms modifying habitats through physical changes. It supports the ecological comparison, not a claim about AI adoption or intention. https://doi.org/10.2307/3545850

Open author-institution copy:
https://www.caryinstitute.org/sites/default/files/public/reprints/Jones_et_al_1994_Organisms_as_Oikos_69_373-386.pdf

[59] Kevin Laland, Blake Matthews and Marcus W. Feldman. **An introduction to niche construction theory.** Evolutionary Ecology 30, 191-202, 2016. Open-access scholarly overview of organisms changing selective environments. Distinguishes niche construction from ecosystem engineering and discusses theoretical disagreements. A different author team, but a related conceptual literature rather than an independent replication of the same experiment.
https://doi.org/10.1007/s10682-016-9821-z

Beyond the Glasshouse: agent-evaluation research [40] and the separate long-task research program [39] motivate scrutiny of practical usefulness. Their settings and measures differ. The proposed paired encounter is an editorial design and has not been run.

The Price of Feathers: problem-dependent benefits of additional computation [36] and joint consideration of cost and accuracy [40] motivate the comparison. No universal trade-off, total cost ranking or new pricing claim follows.

The Habitat Engineers: [58]-[59] supply biological concepts. Human outcomes require their own evidence; [44] and [46] are starting points from different settings. The institutional case remains open. Physical habitat engineering and evolutionary niche construction should not be used as interchangeable technical definitions.

Letters from the Undergrowth: creative editorial fiction. Its incidents, dialogue and first-person voice are invented; no factual product comparison or claim about consciousness is intended.

Visual notes: draft 0.7 replaces several conceptual panels with imagined natural-history scenes and adds illustrations beside other stories. Captions explain the analogy; measured charts remain distinct. The letter now has its own fictional correspondence scene. The front cover remains the approved Illustrated Expedition revision. All twenty new originals and their exact prompts are preserved in the project's illustration folder.

Sources: why evolution is a useful lens

The foundation and two explicit AI mechanisms

These are background papers, not Q4 2026 events. They support specific comparisons rather than a claim that the whole AI industry obeys one biological law.

[60] Richard C. Lewontin. **The Units of Selection.** Annual Review of Ecology and Systematics 1, 1-18, 1970. Foundational analysis of variation, differential contribution to later generations and heredity. Supports the conditions for selection and the need to identify the relevant entities; it does not establish those conditions for modern AI markets. https://doi.org/10.1146/annurev.es.01.110170.000245

Open university-hosted paper: https://www.zoology.ubc.ca/let/pdfs/Lewontin1970.pdf

[61] Alex Mesoudi, Andrew Whiten and Kevin N. Laland. **Towards a unified science of cultural evolution.** Behavioral and Brain Sciences 29(4), 329-347, with commentaries and response through page 383, 2006. Scholarly framework connecting cultural transmission with evolutionary methods. Discusses differences in transmission and deliberate innovation; it is an argument and synthesis, not an AI experiment or proof of universal agreement. https://doi.org/10.1017/S0140525X06009083

Author's page with the full paper and commentaries: https://www.alexmesoudi.com/publication/mesoudi-towards-2006/

[62] Kenneth O. Stanley and Risto Miikkulainen. **Evolving Neural Networks through Augmenting Topologies.** Evolutionary Computation 10(2), 99-127, 2002. Peer-reviewed NEAT paper with experiments on evolving network structures and weights. Its algorithmic grouping called speciation protects structural innovations; it is not a biological species test or a classification of current commercial models. https://doi.org/10.1162/106365602320169811

Final paper hosted by the authors' university: https://nn.cs.utexas.edu/downloads/papers/stanley.ec02.pdf

[63] Max Jaderberg and colleagues. **Population Based Training of Neural Networks.** Research manuscript, 2017. Combines ordinary training with performance-based copying of model settings and variation of training settings. Demonstrates an explicit population process in selected tasks, not the method used by every AI developer. A separate team and method from [62]. https://arxiv.org/abs/1711.09846

Why the pairing matters. [60]-[61] explain the conceptual bridge. [62]-[63] supply two concrete AI mechanisms. Agreement on those mechanisms does not validate our wider claims about product survival, speciation, or the direction of the industry. Those remain subjects for reporting.

Sources: inheritance and technological change

Several ways for an earlier achievement to travel

[64] Edward J. Hu and colleagues. **LoRA: Low-Rank Adaptation of Large Language Models.** International Conference on Learning Representations, 2022; manuscript first posted in 2021. Keeps the underlying model's learned settings fixed while training a smaller set of added adjustments for particular tasks. A concrete route from a shared base to specialized variants; it does not establish reproductive isolation or new biological-style species. https://arxiv.org/abs/2106.09685

[65] Geoffrey Hinton, Oriol Vinyals and Jeff Dean. **Distilling the Knowledge in a Neural Network.** Workshop paper, public manuscript March 2015. Trains a model using information supplied by another model or ensemble, demonstrating transfer through teaching outputs rather than simple copying of all learned settings. Different mechanism and team from [64]; neither paper measures an industry-wide inheritance rate. https://arxiv.org/abs/1503.02531

[66] W. Brian Arthur and Wolfgang Polak. **The evolution of technology within a simple computer model.** Complexity 11(5), 23-31, 2006. A computer experiment constructs logic circuits from earlier components and retains useful combinations. Supports a mechanism of cumulative recombination and dependence on available building blocks. It is a simplified technology model, not an empirical study of AI firms. https://doi.org/10.1002/cplx.20130

Earlier author-hosted working-paper version, dated December 21, 2004:
https://sites.santafe.edu/~wbarthur/Papers/AP-Complexity.pdf

Separate the routes. Retained settings [64], teaching from outputs [65], and component recombination [66] are complementary examples, not three replications of one experiment. ReAct [42] and Toolformer [43] provide additional evidence about composing models with tools. The metaphor must preserve those differences.

No hidden ancestry claims. We do not infer that a particular commercial model is descended from another because their answers look alike. Shared components, copied methods and independently similar solutions require different evidence. Model names and marketing families cannot establish ancestry by themselves.

Sources: feedback, new uses and history

Evidence for the less obvious comparisons

[67] Bowen Baker and colleagues. **Emergent Tool Use From Multi-Agent Autocurricula.** Research manuscript, 2019, revised February 2020. In a simulated hide-and-seek setting, opposing agents developed successive strategies and counter-strategies involving objects and coordination. One bounded experiment showing reciprocal behavioral change during training, not genetic evolution or evidence of an industry-wide arms race. https://arxiv.org/abs/1909.07528v2

[68] Stephen Jay Gould and Elisabeth S. Vrba. **Exaptation - a Missing Term in the Science of Form.** Paleobiology 8(1), 4-15, 1982. Conceptual paper distinguishing present usefulness from the historical origin of a feature. Supports the biological term for a feature recruited to a role it was not originally selected for. Applying it to AI requires evidence of the earlier capability and its new use; a surprising answer alone is insufficient. https://doi.org/10.1017/S0094837300004310

[69] Zachary D. Blount, Christina Z. Borland and Richard E. Lenski. **Historical contingency and the evolution of a key innovation in an experimental population of Escherichia coli.** PNAS 105(23), 7899-7906, 2008. Experiments replayed parts of a bacterial lineage's history and found that earlier changes affected the later emergence of citrate use in oxygenated conditions. This establishes a biological example of historical contingency; it does not quantify technological inevitability. https://doi.org/10.1073/pnas.0803151105

Open paper: https://pmc.ncbi.nlm.nih.gov/articles/PMC2430337/

Evidence across the comparison. Dependence draws on biological hypothesis and experiment [54]-[55], alongside separate tool-use studies [42]-[43]. Habitat change draws on ecology [58]-[59] and workplace or learning evidence [44], [46]; feedback into later AI selection is a proposed investigation. Reciprocal adaptation combines the ecological framework [59] with the limited AI example [67]. Repurposing links [68] with task adaptation [64]. Historical contingency pairs the biological experiment [69] with the technology model [66].

What multiple sources do and do not establish. These are distinct lines of evidence about parts of an analogy, not repeated experimental confirmation of the whole analogy. A biological paper cannot validate an AI product claim. An AI paper cannot establish a biological mechanism. We label the cross-domain connections as editorial synthesis and leave industry-wide generalizations open.

Sources: reading the appetite

Current use, expected use and depth of adoption

[70] U.S. Census Bureau. **Business Trends and Outlook Survey, National.xlsx, October 22, 2026 release.** National Response Estimates, column 202621, questions 7 and 24, Yes rows: 24.6% recent use; 28.6% expected use within six months. Published standard errors: 0.27 and 0.21 percentage points. The chart reports point estimates without error bars. Collection: October 5-18; prior-two-week reference period: September 21-October 4. The dates come from the workbook's Collection and Reference Dates sheet. The survey represents private employer businesses outside agriculture. These are business survey estimates, not consumer usage, spending, or provider revenue. The future response is an expectation, not a realized outcome. The preserved workbook supplies the chart values.

Official data and downloadable tables: https://www.census.gov/hfp/btos/data_downloads

The current questionnaire labels its category "SI (formerly AI)." We use AI for continuity with the journal; this source terminology does not establish that the systems possess superintelligence. The question covers business functions broadly, not just generative AI. We show a snapshot rather than join differently worded questionnaires into an unexplained trend.

[71] Kathryn Bonney, Cory Breaux, Emin Dinlersoz, Lucia Foster, John Haltiwanger and Aditya Pande. **The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks.** Census Center for Economic Studies working paper CES-26-25, April 2026; official page revised October 14. Uses the AI supplement's November 2025-January 2026 reference period. Finds uneven adoption across firms, functions and worker tasks, often with limited breadth. Background research, not the October snapshot. This study and [70] share the Census survey family and must not be presented as independent replications. https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-25.html

[72] Alexander Bick, Adam Blandin, David J. Deming and Tyler R. Schumacher. **What Work Does Generative AI Do?** Research working paper, August 2026; NBER Working Paper 35677. Uses the Real-Time Population Survey, with detailed task evidence from four waves, August 2025-May 2026. Its authors' September 1 summary reports widespread but uneven task-level use. Different respondents and data from the Census research. Self-reported use is not proof of productivity, spending or jobs displaced. The figures describe earlier survey periods, not October outcomes. Paper: https://www.nber.org/papers/w35677

Accessible account by the paper's authors, used to check the findings discussed here:
https://www.stlouisfed.org/on-the-economy/2026/sep/what-work-does-generative-ai-do

Reading the measures together. Firm adoption and worker adoption have different denominators and questions. Do not average their percentages or compare them as if they were one series. Adoption evidence supports a substantial, uneven foothold; it does not establish the size of a profitable worldwide market. Paid retention, spending and supplier margins remain reporting tasks.

Sources: what expansion requires

Complementary investment, energy and conditional futures

[73] Erik Brynjolfsson, Daniel Rock and Chad Syverson. **The Productivity J-Curve: How Intangibles Complement General Purpose Technologies.** American Economic Journal: Macroeconomics 13(1), 333-372, 2021. Explains how investments in organizational changes and other less visible assets can delay measured productivity benefits. A framework for interpreting adoption costs, not evidence that every AI investment eventually pays off. https://doi.org/10.1257/mac.20180386

[74] International Energy Agency. **Key Questions on Energy and AI.** 2026 report, executive summary. Estimates global data-centre electricity use at 485 TWh in 2025 and projects around 950 TWh in 2030. These totals include non-AI data centres. Discusses efficiency, changing tasks and infrastructure constraints. Institutional energy analysis, not a peer-reviewed experiment; projections depend on assumptions and are not promises about realized demand. https://www.iea.org/reports/key-questions-on-energy-and-ai/executive-summary

[75] Alexandra Sasha Luccioni, Yacine Jernite and Emma Strubell. **Power Hungry Processing: Watts Driving the Cost of AI Deployment?** ACM Conference on Fairness, Accountability, and Transparency, 2024. Empirical comparisons of operating energy across selected tasks and models. Supports the importance of workload and model choice; the measurements are not universal energy-per-query estimates for current systems. It does not establish the global size or future growth of AI electricity demand. https://doi.org/10.1145/3630106.3658542

Open conference paper: https://facctconference.org/static/papers24/facct24-6.pdf

How the concepts are supported. Uneven diffusion draws on two research groups and datasets [71]-[72], with [70] providing a newer observation from one survey family. Useful assistance and learning costs connect the worker surveys with the separate workplace study [46] and technology framework [73]. Resource constraints combine energy analysis [74] with empirical task-level measurements [75]; these answer different questions, rather than replicate a global forecast. Human checking and retained skills also draw on [40], [44], [46] and [56].

Our contribution and its limits. The four demand measures and three possible seasons are editorial organizing devices, not established biological laws or forecasts imported from the papers. The watering hole and carrying-capacity language connect demand to the journal's ecological lens. Nobody's surveyed intention is treated as a purchase; announced capacity is not treated as customer demand. No new dated forecast is added to the original sealed-envelope ledger.

Year-end reporting slots. Refresh the latest comparable adoption data; seek evidence of paid retention and spending; examine a documented capacity constraint and a case of sustained value. Record source independence and whether totals include non-AI activity. Do not fill these slots by extrapolating a single provider's growth to the entire ecosystem.